



MATHS in Fashion



Barbara Lynch
Angelo Pantalone



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resources



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Angelo Pantalone

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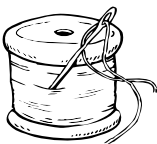
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Maths in Fashion is a collection of reproducible student worksheets which feature the application of mathematical knowledge and techniques to various areas of fashion, textiles and leather. While the content is not necessarily in teaching order, the later worksheets can provide a sequential understanding of the fashion design business.

The local and international fashion, textiles and leather industry employs many millions of people in areas as diverse as the production of raw materials, high-end fashion design, product development and manufacture, marketing, technology, taxes and tariffs, importing and exporting, environmental and sustainability issues. People who study in this area may enter the industry in international trade and global logistics or as producers and retailers, designers and consultants, journalists, stylists, photographers, illustrators, costumiers for theatre and film, or designers of surface and soft furnishings. Mathematics is an integral part of many areas of the industry.

The series has been designed to cover Years 9 and 10 syllabi but the worksheets could be used to motivate and add interest to any junior or middle school Mathematics course. In most cases, an electronic calculator or Excel spreadsheet would be useful. Because the design process is international, units of measurements can be either imperial (inches) or metric (centimetres). Some designers work in inches and then make precise conversions to metric when the patterns are being made.

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The history of fashion is full of interesting ways of dressing and some very funny ideas about what is beautiful or handsome at a particular time in history. Where you see an 'idea' provided for a question, complete the calculations in it to find the answer.

- 1 In what century was it considered fashionable to wear eyebrows that were made from a skinned and cleaned mouse?
Idea: $\frac{1}{5}$ of $10 \times 11 - 4$
- 2 In what year were false eyelashes invented?
Idea: $7 \times 4 \times 68 + 3 \times 4$
- 3 What is the estimated number of pairs of gloves that were owned by Queen Elizabeth I?
Idea: $\frac{4}{5} \times 20 \times 125$
- 4 In what year did the rowing club at Oxford University's Exeter College invent the first school ties?
Idea: $47 \times 2^2 \times 10$
- 5 King Charles VIII of France had trouble finding shoes that would fit him. How many toes did he have on one of his feet?
Idea: $(3 \times 12 + 2 \times 15) / 11$
- 6 How many pairs of gloves were worn at any one time by King Charles VIII's oldest son?
Idea: $(6^2 - 4) \div 2$
- 7 In what year did Marie Antoinette convince the King of France to declare that all handkerchiefs must be square?
Idea: $9 \times 189 + 84$
- 8 How many kilograms of lipstick does the average woman use in one year? Answer correct to 3 decimal places.
Idea: $\frac{3}{4} \times 3.632$
- 9 Indian women have traditionally worn the sari, which is a long piece of silk or cotton that is wrapped over the body in different ways. How long are saris, generally?
Idea: $\frac{225}{45}$ to $\frac{243}{27}$ metres
- 10 Rene Lacoste wanted something more comfortable to wear while playing tennis and so created the Polo 'Tennis' shirt. In what year did this happen?
Idea: $15^2 \times 11 - 2 \times 271$
- 11 In what year was the 'bra' patented by Mary Phelps of New York?
Idea: $190 + 45 \times 20 + 103 + 46 \times 16$

... FAST FASHION FACTS

Order of operations check!

Brackets are first,
Of is second,
Division and Multiplication, then
Addition and Subtraction,
working from left to right.





- 12 a** The largest diamond ever found was called the Cullinan. How heavy was it?

Idea: $6\sqrt{259\,081} + 52$ carats

- b** How much did this diamond weigh in grams, if 1 carat = 0.2 grams?

- 13** How many centimetres below the navel are hipster jeans supposed to start?

Idea: $\frac{10^3 \times 10^2 \times 10}{10^5}$

- 14** How many dollars are spent annually in the perfume industry?

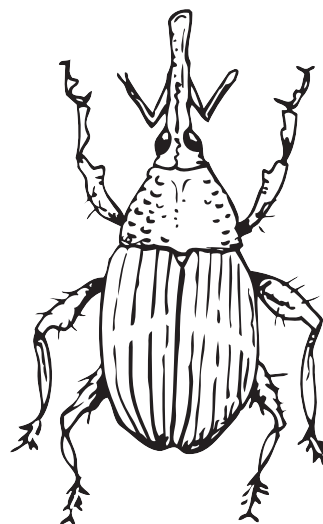
Idea: $\frac{10^5 \times 10^2 \times 10^6}{10^3 \times 10^2 \times 10}$

- 15** Rayon, the first artificial fibre, was invented in 1910 and nylon was invented 29 years later. Aramid fabric was invented 22 years after nylon, and olefin, acrylic, polyester and spandex were invented 12, 11, 9 and 3 years respectively before aramid. List these fabrics in order of invention from earliest to latest and include the year of invention for each one.

- 16** A cotton gin is a machine that can take seeds out of cotton. In what year did Eli Whitney patent his invention of the cotton gin?

Idea: the factors of the year are 2, 3, 13 and 23.

- 17** In Australia, between 200 and 250 million dollars' worth of damage is caused to crops by the boll weevil. This beetle infests the seeds and fibres of the buds of the cotton plant. It causes 5 billion dollars' worth of damage worldwide. What percentage of boll weevil damage is done in Australia? (Hint: one billion = 1 000 million) Give your answer correct to 2 decimal places.



- 18** When jeans are advertised as 'pre-shrunk', it means that the denim has been treated so that the garment will shrink less than 3% in washing. Tegan's new UKOOL pre-shrunk jeans had an 82 cm waist measurement when she bought them. After she washed them, the waist measured 79 cm. If the jeans were really pre-shrunk, is this natural shrinkage?

- 19** Bull denim is a specially woven and dyed fabric made from coarse fibres. The weight of the material can vary from 305 grams per square metre up to the standard 475 grams per square metre. Josh's bull denim jeans used 1.6 metres of 150 cm wide fabric. Complete the following, correct to 2 decimal places.

- a** Find the area of fabric used to make Josh's jeans in square metres (m^2).

- b** Hence find the most that Josh's jeans could weigh in kilograms (kg), excluding the weights of the 195 metres of thread, the 5 buttons and the 6 rivets.

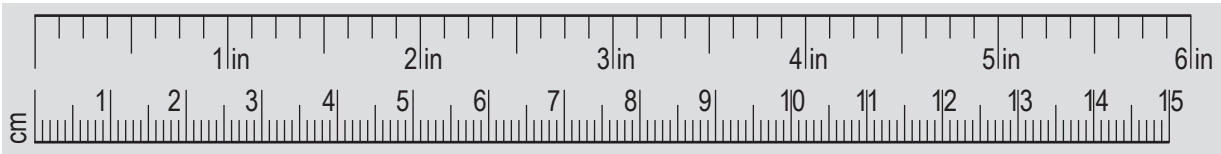
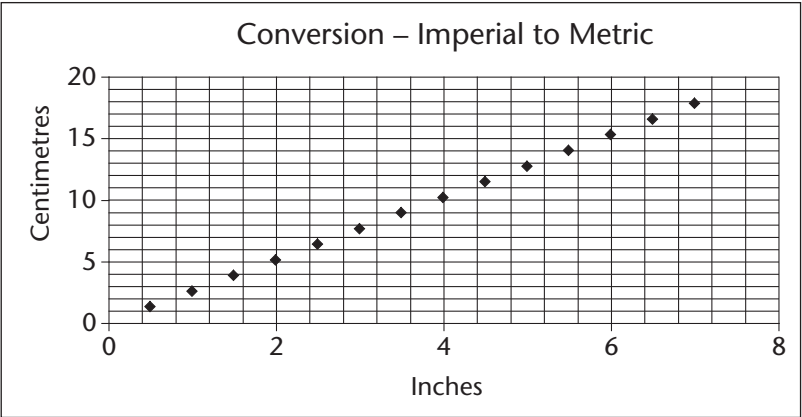
- 20** Use the clues below to find out the price in dollars, P , of a pair of stonewashed, sanded, hand-torn jeans from the Italian design house GRANO-SASSO in 2001.

- P is an even number between 2 000 and 3 000.
- P is divisible by 11.
- All the digits of P are the same.

One of the most popular careers in the fashion industry is the job of fashion buyer. Fashion buyers probably use Mathematics more than most people because the role involves watching sales, developing ranges, negotiating costs and deliveries, and researching the market. Other parts of the fashion industry can involve packing slips, delivery dockets, requisition, quality control, stock management and bills of lading, all of which require mathematical skills.

Items of clothing and footwear come from different countries and so they may not be in the same units. To convert from centimetres to inches or inches to centimetres, you can use one of three methods:

- Use the fact that $2.54\text{ cm} = 1\text{ inch (1")}$.
- Use a tape measure that has both units on it.
- Use a conversion graph.



- 1 UDUDE’s designer decides that a new skirt needs a 4 inch waistband. Help the designer to convert this to centimetres using:
- a the numerical conversion b the tape measure above c the graph above.
- 2 GDONG in China makes shirts for the US market. Convert a 40 inch chest measurement to centimetres using each of the three methods.
- 3 WOBOK runners are marketed internationally. The label shown at the right is attached to each pair of shoes to show the comparison of sizes. Complete the table for WOBOK.
- | Size | Inches | Centimetres |
|------|--------|-------------|
| 7 | 9.875 | |
| 8 | | 25.7175 |
| 9 | 10.5 | |
- 4 Which of these three conversion methods is the most accurate? Why?
- 5 Alex needs new team gear at the start of his footie season with the Hawkeyes. How much will it cost him to buy everything on the list he was given by his club?

HAWKEYES GEAR

- * 1 YOUTH LONG-SLEEVED CLUB JUMPER @ \$84.95
- * 1 CLUB JUMPER NUMBER @ \$10.00
- * 2 PAIRS OF YOUTH PLAYING SHORTS WITH DRAWSTRING @ \$29.95 EACH
- * 2 PAIRS OF YOUTH PLAYING SOCKS @ \$10.95 EACH
- * 1 HAWKEYES T-SHIRT @ \$13.95
- * 1 HAWKEYES CAP @ \$9.95





Hit the shops!

- 6** Lilli has a summer job in a fashion boutique. As a bonus, she is offered 12.5% off the price on the tag of anything in the shop. She has saved up \$85. Can she afford to buy a shirt with a price tag of \$99.99? Why?
- 7** **a** Jon estimated that $\frac{5}{16}$ of his stock of dresses is pink. What percentage of his dresses is pink?
b From the 24 m of racks in the store, Jon needs 60 cm to hang the hoodies. What percentage of the rack will hold hoodies?
c Jon purchases four size 12 dresses for every three size 14 dresses. He bought 35 dresses altogether for the store. How many of each size did he buy?
- 8** What is the new length of Eleanor's skirt after it suffered 7% shrinkage? It was originally 1.4 m long.
- 9** What is the new annual rent at Rod's shop if there was an increase of 3%? The original rent was \$2500 per month.
- 10** For how much did Kath sell pairs of jeans if she bought them for \$45 each and sold them at a loss of 15%?
- 11** What is the selling price of shirts marked at \$36 if Steven was selling them at a discount of 12.5%?
- 12** The sign at the right was used in T Mart's jacket sale.
a How much discount has been allowed?
b What is the discount as a percentage of the marked price?
c If the same percentage discount applied, how much would another jacket marked at \$260 cost?
- 13** Inflation occurs when prices rise and so one dollar buys less. Glowknit shirts cost \$75 three years ago. The inflation rate was 4% then 3% and then 4.5% in successive years. The price of Glowknit shirts rose in line with these inflation figures. Find the price that the shirts will cost at the end of the:
a first year. **b** second year. **c** third year.

... FAST FASHION FACTS

A **discount** is a price reduction, usually calculated as a percentage of the **marked price**. A discount is a *loss* on the marked (original) price.

Discount = marked price – selling price

... FAST FASHION FACTS

- To **increase** a quantity by $r\%$, **multiply** the quantity by $(1 + \frac{r}{100})$.
- To **decrease** a quantity by $r\%$, **multiply** the quantity by $(1 - \frac{r}{100})$.

... FAST FASHION FACTS

A **profit** is made when goods are sold for *more* than they cost. A **loss** is made when goods are sold for *less* than they cost.

Profit (loss) = selling price – cost price

Percentage profit (loss) = $\frac{\text{profit (loss)}}{\text{cost price}} \times 100\%$





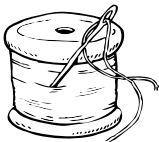
Fashion is a global phenomenon. While certainly there are some countries where clothing, footwear, textiles and leather are just part of everyday life, style and the design of fashion is an industry that continues to play a large part in world trade and employment opportunities.

- 1 'You gotta look cool while you're skateboarding and it can cost quite a bit,' said Mike. He has already bought a pair of new skater high top Reverse kicks for \$124.95, a black CD sweatshirt with dark brown detailing for \$39.50 and Hevi Grouse jeans for \$149.99. His Flexifit cap was \$25 and the black bandana \$9.99. Of course, the most important accessory for a boarder is sunglasses and he is showing off his black-tinted Demon sunglasses that he got on sale for \$50, originally \$149.99.

When he gets enough money he wants to buy the wallet and back pack which will cost him a total of \$57.95. How much will it cost altogether for Mike to be 'skater chic'?



- 2 According to the Organic Trade Association, textile fibre sales are the third fastest growing category of all non-food organic products, with sales of \$160 million in 2005. This increased from \$85 million in 2003. What has been the percentage increase in sales in that period? Answer correct to 1 decimal place.
- 3 The US Government indicated that in 2004 approximately 17 000 people worked as fashion designers. Of these, 25% were self-employed, 25% were clothing wholesalers and 15% were clothing manufacturers.
- How many designers were:
 - self-employed?
 - clothing wholesalers?
 - clothing manufacturers?
 - If the rest of the fashion designers were in management positions, how many were in management?
- 4 According to government records, consumers spent a record \$350 billion on apparel and footwear in a particular country last year. This was an average of \$1 800 for each woman, man and child.
- How many people lived in that country according to this data? (Hint: one billion is a thousand million.) Answer correct to the nearest whole number.
 - These sales from clothing and footwear stores were 6.1% higher than in the previous year. How much did consumers spend in total in the previous year? Answer to the nearest billion dollars.
- 5 The number of fashion designers working in Australia in the years 1991, 2001 and 2006 was counted and found to be 2 054, 2 575 and 2 700 respectively. The data for 1996 was lost, but it is known that the figures for 1991, 1996, 2001 and 2006 average 2 498.75. How many fashion designers were working in Australia in 1996?



Use the fast fashion facts to answer the following questions.

- 6 How many people in Tessuti are employed *in some way* in the textile, clothing, footwear and leather industry?
- 7 If the population of Tessuti is 208 000, what proportion of people are employed in some way in this industry?
- 8 What is the average salary of those *directly* employed in this industry?
- 9 Turnover is the total amount of money received before deducting any costs. How much does each directly employed person contribute to the turnover?

... FAST FASHION FACTS

- According to records in the regional city of Tessuti, the textile, clothing, footwear and leather industry directly employs 2 200 people with a turnover of \$440m and salaries of \$70m annually.
- In addition to this, more than 300 people are employed in research, education and training in this sector.

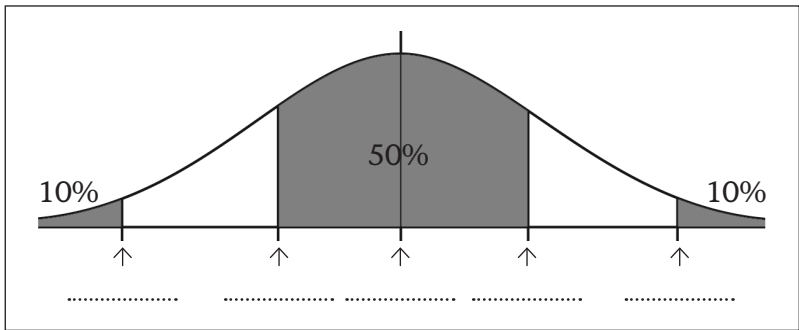
Use the fast fashion facts about the fashion industry in the USA to answer the following questions.

- 10 How many fashion designing jobs are there expected to be in the next 6 years?
- 11 How many fashion designers will still be in the business after 2 years?
- 12 How many fashion designers will have dropped out after 5 years?
- 13 How much per hour does the average fashion designer earn for 52 weeks per year of work?
- 14 Complete the diagram below by filling in the correct salaries on the dotted lines.

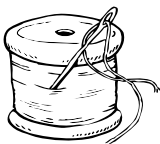
... FAST FASHION FACTS

Information collected from 20 000 people employed as fashion designers indicated that they think:

- the number of designers will increase by between 10 and 12% in the next 6 years
- more than 92% will still be in the business after working 2 years
- almost one half will leave the business after working 5 years
- the average fashion designer works 55 hours per week
- the average fashion designer earns \$62 610, the middle 50% earns between \$42 140 and \$87 510, the lowest 10% earns less than \$30 000, and the highest 10% earns more than \$117 120.



- 15 The graph in question 14 is symmetrical and the total area underneath it represents 100%. Use it to find the earnings of those in the:
 - a lower 25%.
 - b upper 25%.
 - c lower 90%.
 - d upper 90%.



People entering the fashion field soon find that they need a range of skills as well as a good understanding of Mathematics. Fashion is both local and global, both an art and a business.

The following table shows the different types of designers in each state and territory of Australia expressed as a percentage of the total number of designers. Use this data to answer the questions below.

Percentages of different types of designers by state or territory of Australia

Type of designer	SA	WA	VIC	NSW	QLD	ACT	TAS	NT	Total
Graphic designers	4.06	4.69	17.47	22.81	10.22	1.71	1.36	0.42	62.75
Industrial & product designers	0.23	0.36	1.42	1.36	0.66	0.08	0.01	0.00	4.13
Interior designers	1.34	2.29	5.84	9.54	4.25	0.83	0.46	0.13	24.68
Textile & fashion designers	0.09	0.30	1.02	1.23	0.53	0.08	0.06	0.01	3.32
Interior decorators	1.66	0.32	1.07	1.29	0.56	0.08	0.06	0.09	5.13
Total	7.38	7.97	26.81	36.23	16.23	2.78	1.95	0.65	100.00

- 1

Which state has the greatest number of:

a

textile and fashion designers?

b

interior decorators?
- 2

a Which design job is the most popular?

b If there are 9 950 designers in Australia, how many have the most popular design job?
- 3

In which state are there the fewest industrial designers?
- 4

In which territory are there the fewest interior decorators?

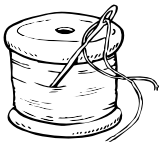


Put your best foot forward!

- 5

A new brand of skate shoe called FAST... is entering the market. The target market is skate-savvy boys aged 15–19 years. The designer wants to sell in Australia, the UK and the US, and the sizing is very different in these markets.

System		Sizes															
Europe		35	35½	36	37	37½	38	38½	39	40	41	42	43	44	45	46½	48½
UK	M	3	3½	4	4½	5	5½	6	6½	7	7½	8	8½	10	11	12	13½
	W	2½	3	3½	4	4½	5	5½	6	6½	7	7½	8	9½	10½	11½	13
Australia	M	3	3½	4	4½	5	5½	6	6½	7	7½	8	8½	10	11	12	13½
	W	3½	4	4½	5	5½	6	6½	7	7½	8	8½	9	10½	11½	12½	14
US & Canada	M	3½	4	4½	5	5½	6	6½	7	7½	8	8½	9	10½	11½	12½	14
	W	5	5½	6	6½	7	7½	8	8½	9	9½	10	10.5	12	13	14	15.5
Centimetres		22.8	23.1	23.5	23.8	24.1	24.5	24.8	25.1	25.4	25.7	26.0	26.7	27.3	27.9	28.6	29.2



- a A Canadian skater, The Torpedo, needs a size 12½. What shoe size would he take in an Australian fit?
- b FAST... has decided to introduce a shoe collection for young females (15–19 years) for ‘street dance’, branded as GLAM. The US has launched a size range from 22.8 cm to 25.4 cm. Complete the table to show the shoe sizes for the US market.

Centimetres	22.8	23.1	23.5	23.8	24.1	24.5	24.8	25.1	25.4
US sizes									

- c The UK market has problems selling the two smallest sizes in the GLAM collection and needs the shoes to start at 23.5 cm. Nine sizes are needed to form a suitable collection. Complete the table to show the sizes of the UK female range.

Centimetres	23.5								
UK sizes									

- d Missy Glide, a street dance artist in New York, has a foot that measures 25.7 cm. What size would she have in the US?

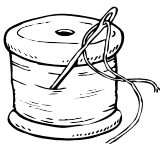


- 6 TLX is a fashion clothing shop. The owner Thea is concerned that her monthly expenses are getting out of hand. Her expenses are made up of two types: fixed expenses that she always has to pay to stay in business, and variable expenses that can change and depend on how much business she is doing.
- She used the tables below to list all her expenses for one month and worked out the totals.

- a How much does she spend in total on fixed expenses per month?
- b How much does she spend in total on variable expenses per month?
- c What percentage of her total expenses is fixed?
- d What percentage of her total expenses is variable?
- e She decides to reduce advertising and travel expenses by half. How much will she save?

Fixed expenses	€ per month
Rent	3 575.00
Insurance(s)	1 400.00
Rates	540.00
Fixtures and fittings	1 456.00
Duty (if applicable)	3 500.00
Taxes	3 240.00
Equipment (computers, etc.)	450.00
Freight	85.00
Total	14 246.00

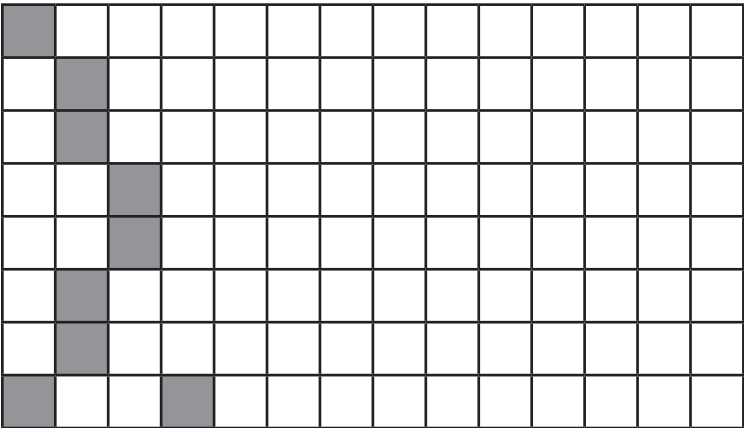
Variable expenses	€ per month
Wages including superannuation, loading and work cover	9 500.00
Power	155.00
Advertising	4 500.00
Customer Returns	4 680.00
Travel (Local + International)	450.00
Office Supplies	1 850.00
Store Thefts (called ‘shrinkage’)	580.00
Marketing	4 400.00
Total	26 115.00



The patterns and shapes in textile designs vary between cultures. For example, there are the Maori diagonals, Hmong spirals, Mayan spiny stars, Navajo eye-dazzlers and Amish four-patch designs. Some patterns have special symmetry. Many are made by repeating shapes over and over again to cover a space without leaving any gaps or having any overlaps. These are called tessellations. Often traditional designs play a significant role in the development of modern fashions.

Maori diagonals

- 1 The diagonal pattern started below is to cover the whole 140 cm by 80 cm panel.
 - a Complete the pattern for the whole panel.
 - b Find the total area of grey squares on the panel.
 - c Find the area of the panel not covered by grey squares.
 - d A new panel is to be made with the same width of 80 cm but a different length. How long will the panel be if the basic pattern repeats 6 times?

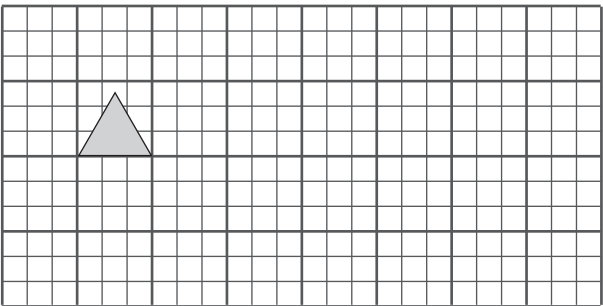
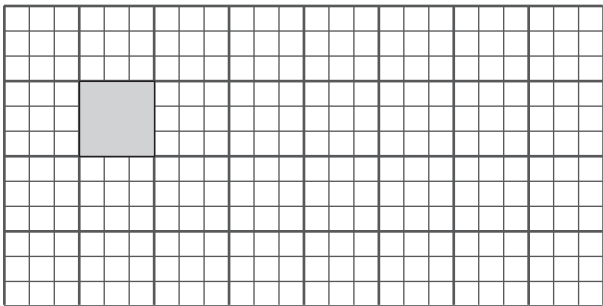


... FAST FASHION FACTS

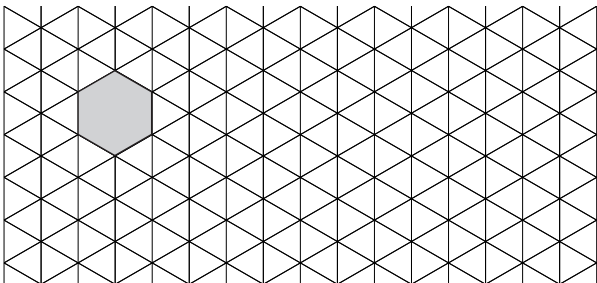
Tukutuku textile designs are used on woven panels in meeting houses. They are usually designed to illustrate a story.

Tessellations and tiling patterns

- 2 Use the graph paper to show how these shapes will repeat to fill a space:
 - a 10 squares
 - b 18 triangles



- c 10 hexagons



... FAST FASHION FACTS

- Only three regular polygons tessellate: triangles, squares and hexagons.
- Regular means sides of equal length.
- Regular tessellations are made up of identical regular polygons.



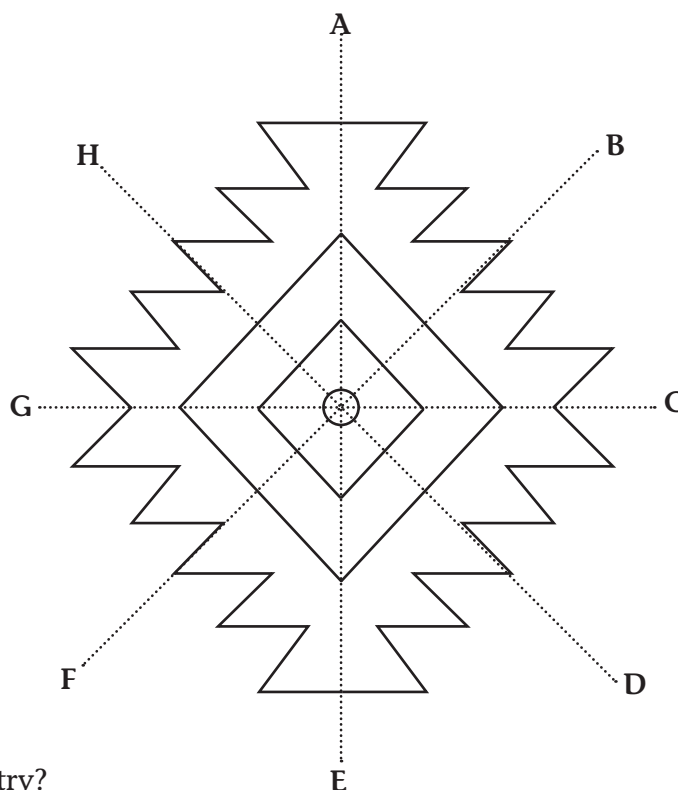
Mayan spiny stars

... FAST FASHION FACTS

The Mayan Indians weave the same designs today as they did in the 1st century. This spiny star design is a symmetrical figure usually found on the clothing of an important person in the tribe.

Symmetry

- A shape has **line symmetry** when one half is the mirror image of the other half.
- A shape has **rotational symmetry** of order 3, for example, when it matches itself three times during one rotation about a centre point.



- 3** For the Mayan spiny star figure shown:
- which lines are lines of symmetry?
 - what is the rotational order of symmetry?
That is, when you rotate the figure about the centre, O , how many times does the figure match itself?

Amish quilts

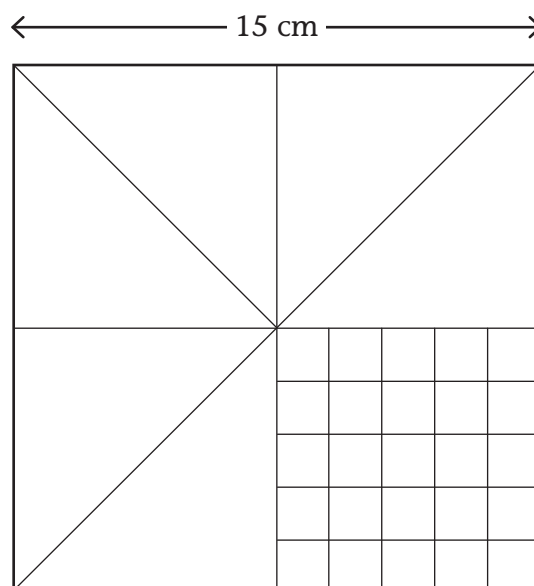
A quilt is a type of doona made up of three layers of material sewn together. The top layer is made up of a grid of scraps of material squares and triangles sewn together.

The diagram shows a 'four-patch block' and these four square patches will be the unit repeated throughout the quilt. Each four-patch block is 15 cm wide.

... FAST FASHION FACTS

The Amish people are a religious group which began in 1693 and is centred around Pennsylvania, USA. The men's dark coats and broad-brimmed hats and the women's long dresses and bonnets reflect that era. Amish handmade quilts are known for their simple and striking designs and solid colours.

- How wide is each of the four square patches?
- The four-patch blocks are to be used for a quilt for a double bed. The dimensions of the quilt will be 150 cm wide and 180 cm long.
 - How many four-patch blocks will be needed?
 - How many square patches will be needed?
- Find the total area of material in the three layers sewn together, in m^2 .
- Colour the four-patch block. Note that the Amish never use white because of its association with death.





The patterns and shapes in textile designs vary between cultures. For example, there are the Maori diagonals, Hmong spirals, Mayan spiny stars, Navajo eye-dazzlers and Amish four-patch designs. Some patterns have special symmetry. Many are made by repeating shapes over and over again to cover a space without leaving any gaps or having any overlaps. These are called tessellations. Often traditional designs play a significant role in the development of modern fashions.

Navajo eye dazzlers

1 At an auction of 'eye-dazzler' weavings, these items were sold: 8 pillow covers measuring 60 cm^2 for \$280 each, a 43 cm^2 woven picture for \$550 and a sampler measuring 40 cm by 54 cm for \$350. A blanket measuring 270 cm by 210 cm was very expensive at \$15 500.

a How much was paid in total for the Navajo weavings?

b Which was the most expensive weaving in cents per square centimetre?

2 The auctioneer said: 'The price of such weavings has risen between 10 and 40% in less than two years. During the last decade (10 years), they have increased 10 times or more in value.' To the nearest whole number, how much would one of the pillow covers have been worth:

a 2 years ago?

b 10 years ago?

... FAST FASHION FACTS

During the late 1800s, Navajo women wove visually striking blankets and rugs called 'eye-dazzlers'. These designs used colours such as the brilliant red, green, yellow and blue.

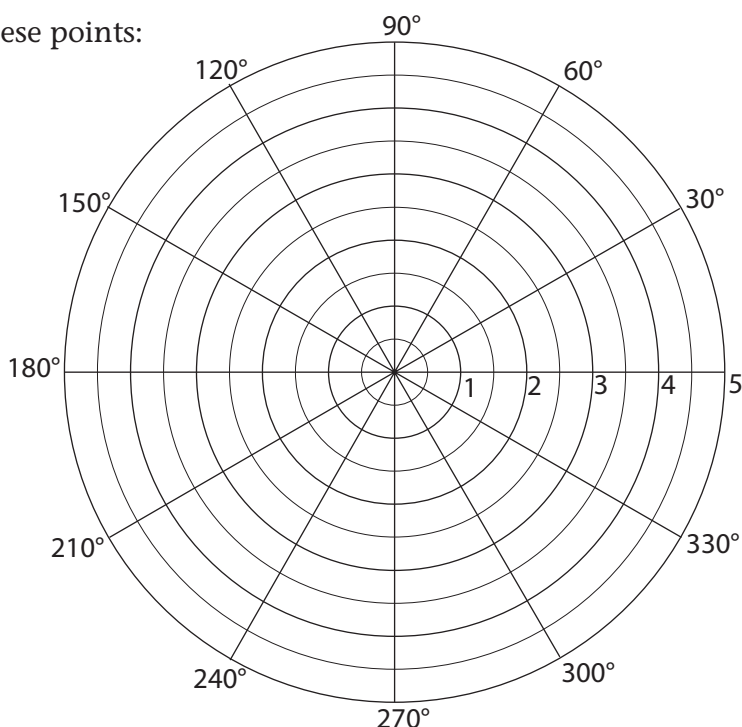
Hmong spirals

A spiral is a curve in the plane that runs around a centre. There are many types of spirals. The easiest way to draw them is to plot points on special graph paper called polar graph paper.

Each point is marked according to its radius from the centre (0, 1, 2, 3, 4, 5) and its angle around the centre (0° , 30° , 60° , 90° , ... , 360°).

3 Use the polar graph paper to plot these points:

- [3, 60°]
- [1, 150°]
- [4, 210°]
- [2, 330°].

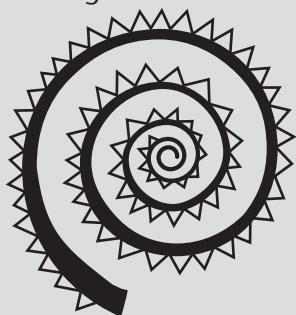


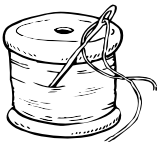
... FAST FASHION FACTS

The Hmong are an ancient people who live in parts of Laos and Vietnam.

Paj ntaub or 'flower cloth' is their traditional way of decorating clothing.

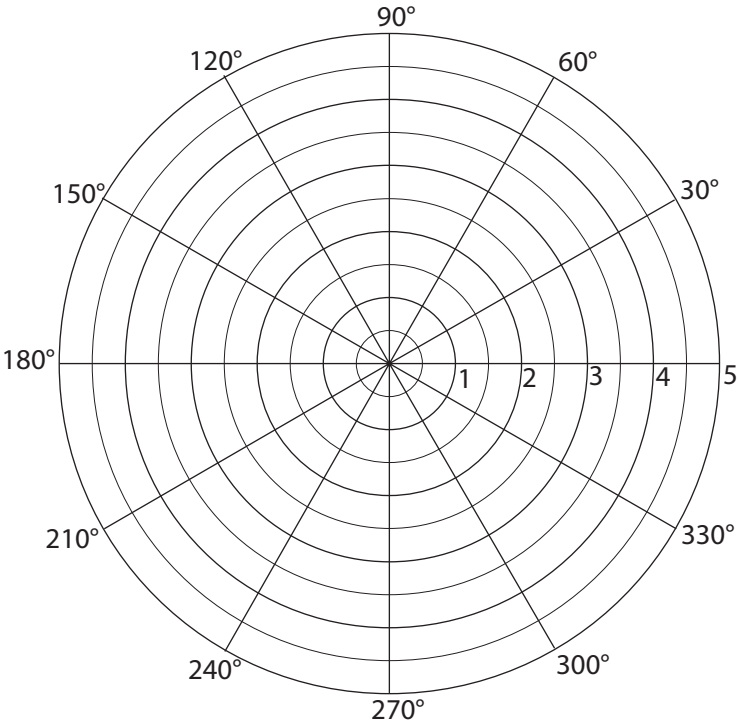
The Hmong spiral shown here represents a dragon's tail.





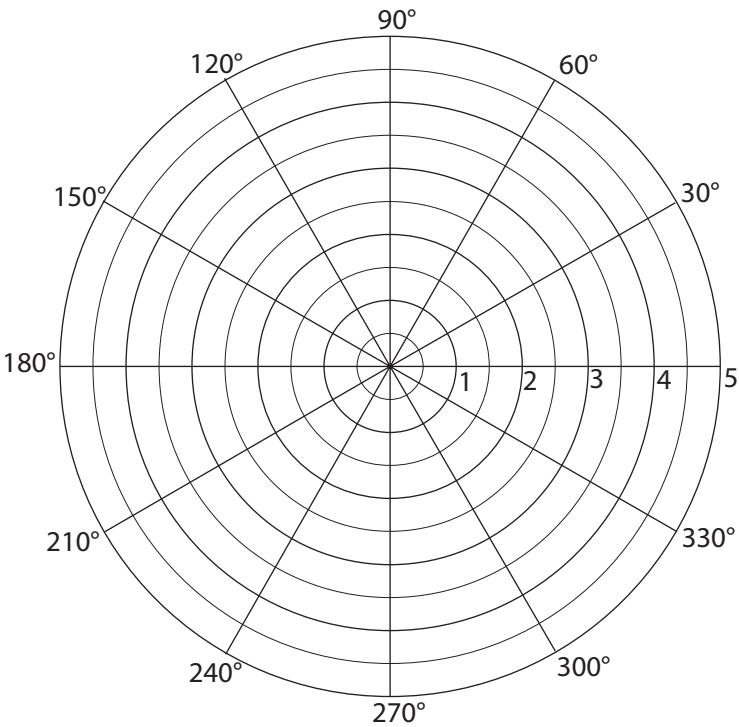
4 Archimedes' spiral: Use the polar graph paper to plot the points from the given table of values showing radius (0, 1, 2, 3, 4, 5) and angle (0° , 30° , 60° , 90° , ... , 360°). Join the points to form a spiral.

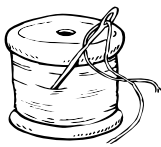
Radius	Angle	Point
0	0°	$[0,0^\circ]$
1	60°	$[1,60^\circ]$
2	120°	$[2,120^\circ]$
3	180°	$[3,180^\circ]$
4	240°	$[4,240^\circ]$
5	300°	$[5,300^\circ]$



5 Equiangular spiral: Use the polar graph paper to plot the points from the given tables of values showing radius (0, 1, 2, 3, 4, 5) and angle (0° , 30° , 60° , 90° , ... , 360°). Join the points to form a spiral.

Radius	Angle	Point
0	0°	$[0,0^\circ]$
0.07	60°	$[0.07,60^\circ]$
0.2	120°	$[0.2,120^\circ]$
0.58	180°	$[0.58,180^\circ]$
1.65	240°	$[1.65,240^\circ]$
4.7	300°	$[4.7,300^\circ]$





The clothing industry is the second largest industry in the world, after telecommunications. At the same time it is the third largest polluter in the world. Tomorrow's fashions may be made from new and exotic materials, may contain bionic devices or be antibacterial. But they will also need to be kind to the planet – that is, eco-friendly. Eco-fashion is about making clothes that care for the environment, the consumers and the working conditions of the people in the industry.

- 1 Handbags made from recycled lolly and chip wrappers sell for \$28 to \$58. It is expected that sales of these bags next year will be 1.5 million. What will be the value of next year's sales?
- 2 Research has shown that 454 g of fertiliser and pesticides are needed to grow the 1.362 kilograms of cotton needed for a T-shirt and a pair of jeans. Answer the following correct to 2 decimal places.
 - a How many T-shirts and pairs of jeans can be made from 100 kg of cotton?
 - b How many kilograms of fertiliser and pesticides are needed to grow 100 kg of cotton?

... FAST FASHION FACTS

You can now buy polar fleece jackets made from 70% organic cotton and 30% recycled plastic bottles.

A group of students was given a list of effects of the fashion industry on the environment and asked which effect was the most important to them. The results of the survey are shown below.

Effect on the environment	Number of students who felt this was the most important effect	Fraction of students who felt this was the most important effect	Number of degrees on pie chart
Pesticides used by farmers	432	$432 \div \text{total number surveyed} = 0.3$	$0.3 \times 360 = 108$
Chemicals used to bleach, colour and print	576		
Textiles and clothing ending up as landfill rather than being recycled	288		
Poor conditions for workers making clothing	144		
Total number surveyed	1440		360

- 3 Use the table of results to find:
 - a how many students were surveyed.
 - b which effect on the environment the students felt was most important.
- 4 Calculate the fraction of students who found each effect the most important and complete the third column of the table.
- 5 To put the information into a pie chart, you need to calculate the size of the angle representing each effect. Complete the last column of the table.
- 6 Hence draw the pie chart.





Future fabrics

7 Kevlar® is an artificial fibre that was invented in 1965 in DuPont's laboratories. It is extremely strong, lightweight, flexible, flame-resistant and five times stronger than steel. It is used for protective clothing and bulletproof vests.

- a** Find the area of Kevlar fabric needed for a jacket if it is 1.27 m wide and 2 m long.
- b** Find the weight of the fabric needed for the jacket if 1 m² of Kevlar fabric weighs 170 g.

8 A new sweat-absorbing fabric called Derasilk is going to be used for socks. The sock designers need to know how much sweat the average person produces in their feet. The average person walks about 10 000 steps every day.



- a** If each step averages 60 centimetres and a lifetime of walking averages 65 years (of 365 days), how far in kilometres will a person walk in a lifetime?
- b** Earth has a circumference of 40 076 km. How many times around the Earth will the average person walk in a lifetime? Give your answer as a whole number.
- c** If each of the 125 000 sweat glands in a foot produces 0.002 mL of sweat per day, how much sweat do two feet produce in one day?
- d** A pair of cotton socks weighs 45 g and absorbs 10% of its weight in sweat without feeling damp. A pair of Derasilk socks weighs 38 g and absorbs 30% of its weight in sweat. If 1 mL of sweat weighs 1 g, find:
 - i** the number of mL of sweat a pair of cotton socks will absorb.
 - ii** the number of mL of sweat a pair of Derasilk socks will absorb.
- e** Use your answer to part c to find if either cotton or Derasilk socks will absorb a day's sweat.

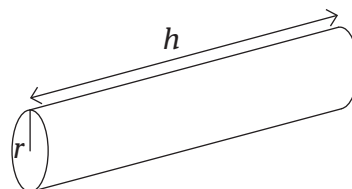
... FAST FASHION FACTS

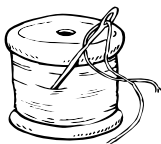
Nano fabrics can sense the temperature of the environment, massage the feet, resist bacteria, be sweat and dirt resistant and be easy to clean. Great for school uniforms!

9 A nanometre is $\frac{1}{1\,000\,000\,000}$ metre. This is the same length as 3 atoms in a row. Graphene is the thinnest possible fabric and is only one atom thick.

- a** How many nanometres thick is graphene?
- b** If a human hair is approximately $\frac{1}{10\,000}$ metre, how many sheets of graphene would be the same thickness as a human hair?
- c** Nanotubes are rolled-up graphene strips that are 1 500 nanometres long and less than 1 nanometre in diameter. If you assume that the tube has a circular cross section as shown, what is the volume of the nanotube in cubic nanometres?

Hint: Volume of a cylinder = $\pi r^2 h$.





Choosing the right size clothing can be rather difficult as you need to know what units of measurement and what country's sizing system are being used. Many designers still use inches rather than centimetres and then have to convert the measurements back. Finally, the measurement of the actual clothing depends on whether the clothes are designed to be a tight fit or a loose fit. Fashion trends usually determine the fit of clothing for a particular season.

Female - What size am I?

1 The following table shows a sample female size chart like those used by designers and retail stores in Australia and Asia. European sizes are important, as Australian and Asian stores also sell European merchandise. Use the table and the fast fashion facts to answer the following questions.

Standard size	6	8	10	12	14
to fit Bust (cm)		75	80	85	
to fit Waist (cm)		55	60	65	
to fit Hip (cm)		80	85	90	
European size	36	38	40	42	44

- a Continue the pattern to fill in the blank measurements for bust, waist and hip sizes 6 and 14 in centimetres.
- b What is an Australian model's standard waist measurement in centimetres?
- c Convert the standard size 12 bust measurement into inches.
- d Michaela has a waist measurement of 24 inches. What standard size does she take?
- e Tegan is an Australian house model. Calculate her bust, waist and hip measurements in inches. Answer correct to the nearest half inch.
- f Sophia is an Italian house model. What is her hip measurements in centimetres?
- g Italian swimwear brand Aqua Bella is launching a new bikini line at the Milan fashion show. Find the bust measurement in centimetres of Lola the swimwear house model.

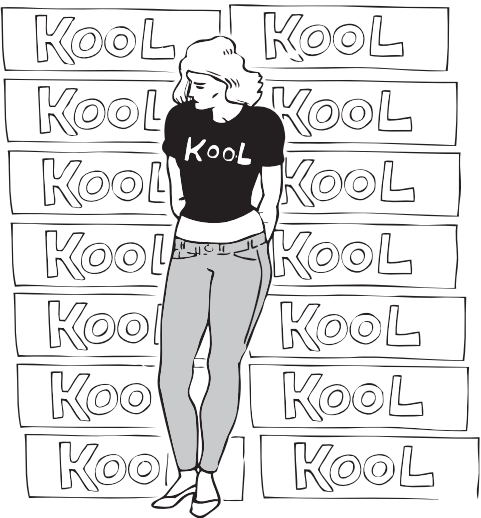
... FAST FASHION FACTS

Australia has **standards** that should be used when developing clothing for females and males, including children (AS1344-1975).

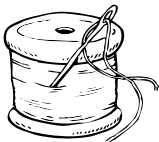
- Australian female sizes were established in **1959**.
- Australian female house models are usually a **standard size 10**.
- European female house models are usually a size **40**.
- Lingerie or swimwear house models are often **size 10** but a **size 11** (halfway between 10 and 12) in the bust.
- Conversion: **1 inch = 2.54 cm**

Do I look big in these jeans?

- 2 A retailer has developed a jean brand called KOOL. In order to attract more customers, they have decided to manufacture half sizes from sizes 6 to 14. This means having a size 7 between size 6 and size 8, a size 9 between size 8 and size 10 and so on.
- a Complete the table of waist sizes in centimetres.



Size	6	7	8	9	10	11	12	13	14
Waist measurement	50 cm		55 cm		60 cm		65 cm		70 cm



- b The Marilyn style of jean sits low on the hip. Tegan the house model needs an extra inch (2.54 cm) around the hips so the jeans fit better. She usually takes a standard size 10. Use the size chart table in Question 1 to find Tegan’s new hip size in centimetres.
- c Convert Tegan’s hip measurement into inches, rounded to the nearest half inch.

Male - How do I buy my first suit?

- 3 Male sizes are given in metric, imperial and European sizes. The following size chart table shows chest and waist sizes. Refer to it to answer the following questions.
- a The metric sizes are not exact compared with the imperial inches measurements. Use the fact that 1 inch = 2.54 cm to complete the table with the exact values in centimetres, correct to 2 decimal places.

... FAST FASHION FACTS

Today’s male suits are fashioned on military suits of the past.

Male house models are usually a size 40 inch chest and 34 inch waist.

Chest	Imperial (inches)		36	38	40	42	44
	Metric (cm)		92	97	102	107	112
	European sizes		46	48	50	52	54
	Exact values (cm)		91.44				
Waist	Imperial (inches)	28	30	32	34	36	38
	Metric (cm)	72	77	82	87	92	97
	Exact values (cm)	71.12					

... FAST FASHION FACTS

When making male suits, designers subtract 6 inches from the chest size to get the waist measurement. This is referred to as the **drop 6 principle**.

- b Use the fast fashion facts to find your waist measurement in inches, if your suit was a standard size 38 inch chest.
- c If I measure 92 cm in the waist, find the jacket size I should try on in whole centimetres.
- d If I purchased a suit in Paris sized 52, what would be the pant waist size in centimetres?
- e Jimmy measures 30 inches in the waist and wants to buy a European suit brand called SMITH & CO. What would Jimmy’s European chest size be?



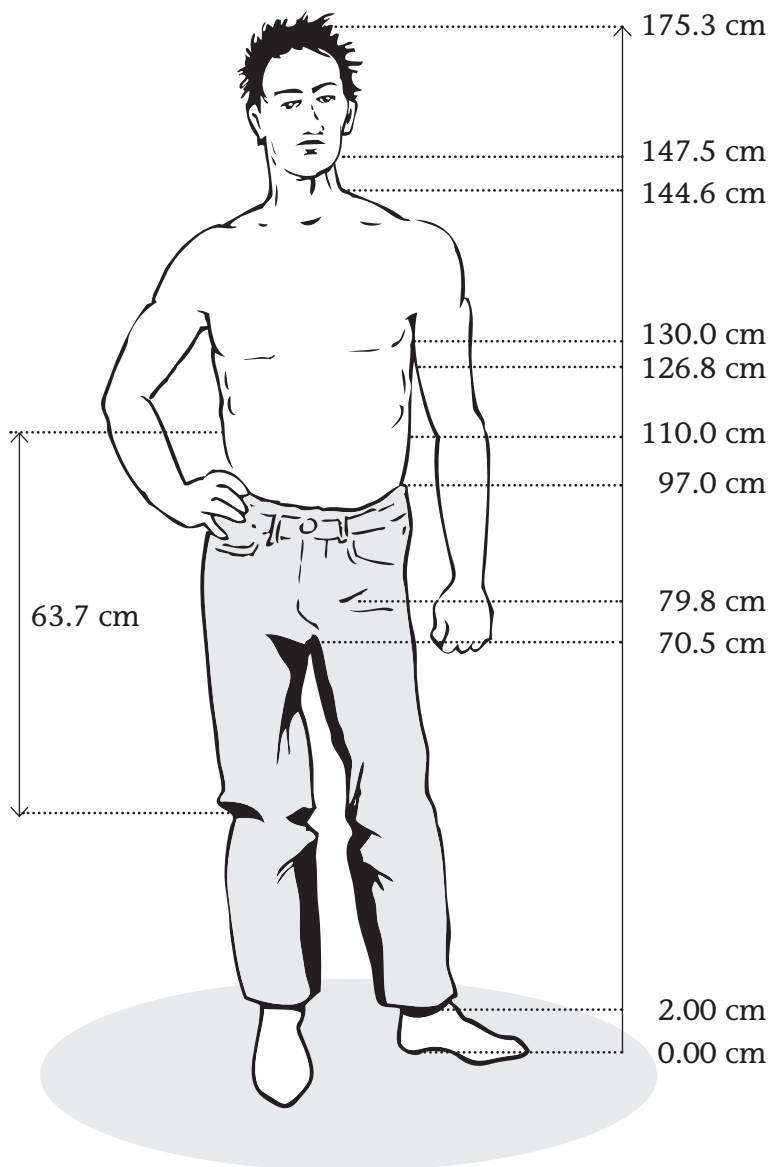


A body scanner takes a three-dimensional image of a person from which all the body measurements can be found automatically. These measurements can then be sent via the Internet for people to make patterns for new clothes or plastic mannequins for displays. Such scans can also help to make standard charts of measurements for designers to use. Of course, measurements can vary between brands and cultures and often the whole process of design is completed in inches rather than centimetres.

Here is Alex's body scan. Use it to answer the following questions.

- 1 How tall is Alex?
- 2 How far is it from the top of his neck to the top of his head?
- 3 How long is his neck?
- 4 How far is his knee above the floor?
- 5 How far is his waist above his belt?
- 6 How far is his crotch below his buttocks?
- 7 If his shirt is to be long enough to go from the bottom of his neck to his crotch, how long must it be?
- 8 The inside leg measurement is from the crotch to the bottom of the trousers. What is Alex's inside leg measurement?
- 9 The girth (circumference) of the middle of Alex's neck is 40.3 cm and at the base of his neck it is 44.9 cm. What is his collar size in whole centimetres if he takes the average of these two figures?
- 10 Alex's chest girth is 109.1 cm and his maximum belly circumference is 91.5 cm. If he buys a suit:
 - a what chest size should he buy, 42" or 44"?
 - b what waist size should he buy, 36" or 38"?

(Hint: Remember that there are 2.54 cm in 1".)



Will my gym work give me a fat neck?

- 11 Use the pattern of measurements in the table below to complete:
- a the row of chest measurements ending in size 50"–52".
 - b the rows of neck and sleeve measurements.

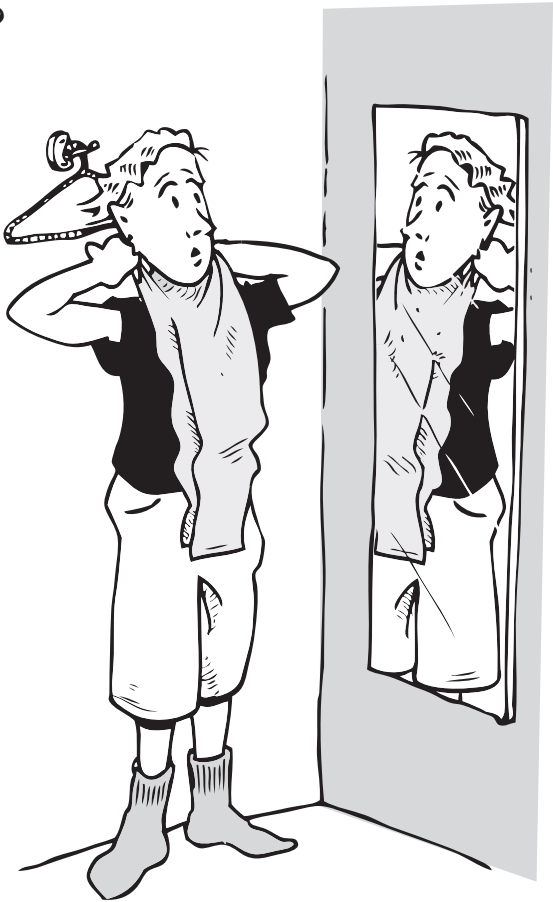
Men's body measurements

Chest	34"–36"	38"–40"			
Waist					
Neck	14½"–15"	15½"–16"			
Sleeve	32"–33"	33–34"			

- 12 When making men's suits, designers subtract 6 inches from the chest size to get the waist measurement. This is referred to as the 'drop 6 principle'. Use this principle to complete the row of waist measurements.
- 13 Eric's gym workout has resulted in his neck increasing to 18". His sleeve length is normally 34". Will the shirt he buys to fit his neck have the correct sleeve measurement? If not, will the sleeve be too long or too short?
- 14 Brad works out at the same gym as Eric and has managed to trim his waist to 32". His chest is 40". Does the 'drop 6 principle' work for him now? Can he wear a ready-made suit or will he need to have it altered?

Fashion folklore or fashion fact?
What's my neck size got to do with my waist size?

- 15 A size S man's chest measures 36 inches. Using the 'drop 6 principle' described in question 12, what would be his waist size?
- 16 From size charts, it appears that, depending on the culture, the neck size is half the waist size. Rodney held up a zipped-up pair of pants and placed them around his neck to see whether the pants would fit him. He knows that his neck measures 17 inches. What would Rodney measure:
- a around the waist?
 - b around the chest?





There are three primary colours: red, yellow and blue. These colours can be mixed together to give secondary colours: red and yellow make orange, red and blue make purple, and blue and yellow make green. Artists use colours in this way. Computers use a system based on red, green and blue. Printing uses cyan, magenta, yellow and black.

It was difficult to describe the approximately 1900 colours available to designers and so the Pantone® Matching System was created in 1963. Each of the standard colours is given a number such as PMS 194. This number can be used for the fabric, the computer screen or the printed brochure without any confusion.

- 1 Mauve was the first colour that was not made from insects or plants, and could be mass-produced in a factory. In what year did William Perkin accidentally make mauve while he was looking for a treatment for malaria? How old was he then?

Hint: $\frac{\text{year}}{\text{age}} = 103\frac{1}{9}$ and year + age = 1874

- 2 So that the designer can see all the colours easily, Pantone guides often consist of a number of cardboard strips, each printed with related colours. In one particular colour guide there are 7 shades per blade. How many blades are there in this guide if there are 1 925 textile colours?

- 3 De Milo Designs need 60 different colours for the autumn season. To make sure that everyone is working with the same colours, they dye 10 cm by 10 cm pieces of cotton with each colour. These squares are called 'swatches'. They cut each swatch into 2 cm squares so they can send a complete set of the colours to each of their suppliers.

a How many suppliers can be provided with a complete set of colours?

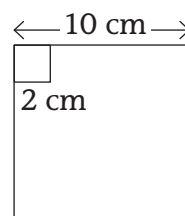
b How many 2 cm squares of cotton will they be handling in order to do this?

... FAST FASHION FACTS

PMS colours are used in branding so that logos and advertising all show exactly the same colours.

A country's flag will have its own pantone numbers.

A computer screen does not accurately show some colours.



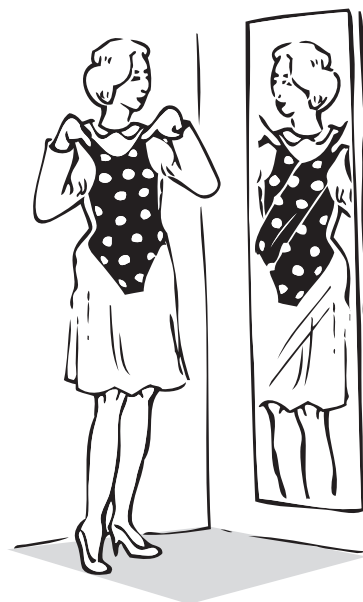
During the last summer season of 12 weeks, De Milo Swimwear sold 1 200 pairs of bathers on the Internet. Each pair cost \$18.00, and postage and handling of \$4.50 per pair was added to the price.

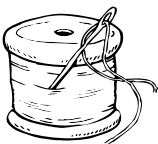
- 4 If 12% of those sold were returned because the colour was not what the customer expected, how much money did De Milo receive for bathers:

a for the season?

b per month?

- 5 De Milo has been offered software that will make sure that the colours the customer sees on the computer screen are 'what-you-see-is-what-you-get'. If this software costs \$1 500 per month, is it worth their buying it to make sure that there are no returns due to wrong colour?





- 6 According to research, 60% of Internet shoppers do not trust the colours they see on screen and in fact will not buy some items online because of this. They say they would buy if the colours were guaranteed to be correct. De Milo Swimwear does guarantee their customers will receive the colour they see on the screen. During the next month, of all the people who visit the website, 12 500 visits become actual sales.
- a How many bathers do De Milo expect to sell?
 - b How much money will they receive in the month?
 - c If they paid \$25 000 for advertising the guarantee, how much will they receive once this cost has been subtracted?
- 7 If the PMS colours had been described by 3-digit numbers only from 100 to 999, how many colours could have been described?
- 8 Digital cameras and computers use a different set of primary colours: **Red**, **Green** and **Blue**. This is called the RGB system. These colours can be mixed together to give secondary colours: green and blue make cyan, red and blue make magenta, and red and green make yellow.
- a If there are 256 different shades of red, 256 shades of green and 256 shades of blue (each going from 0 to 255), how many colours are possible under the RGB system?
 - b Use the fast fashion facts to find out which colour is (0, 0, 0).
 - c What colour is (255,255,255)?
- 9 Most modern colour printing systems use the CMYK process, which has four different coloured inks: **Cyan**, **Magenta**, **Yellow** and **black**. The UBEAUT team has chosen the colours for their new range of scarves: PMS 178, 218 and 803.
Use the table below to answer the following questions.

... **FAST FASHION FACTS**

RGB is called an **additive system** because you start with **black** and **add** colours.

A purple colour called magenta would have a mix of 128 parts red, 0 parts green and 128 parts blue.

Another purple has a more greenish tone if it is a mix of 255 parts red, 64 parts green and 255 parts blue.

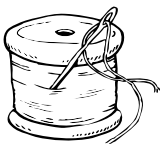
... **FAST FASHION FACTS**

CMYK is known as a **subtractive system** because you start with **white** and **subtract** the colours.

Black is needed because cyan, magenta and yellow mix to make a dirty brown, not a perfect black.

PMS	RGB	CMYK	Description of colour
100	(244,237,124)	(0.000, 0.029, 0.492, 0.043)	Buff
178	(249,94,89)	(0.000, 0.622, 0.643, 0.029)	Red-brown
218	(237,114,170)	(0.000, 0.519, 0.283, 0.071)	Lolly pink
300	(0,114,198)	(1.000, 0.424, 0.000, 0.224)	Blue
724	(142,73,5)	(0.000, 0.486, 0.965, 0.443)	Brown
803	(255,237,56)	(0.000, 0.071, 0.780, 0.000)	Golden yellow

- a What are the names of colours to be used in the scarf range?
- b How will these colours be described for the computer screen with the RGB system?
- c Look at the format of the CMYK system. How many colours are possible? Give your answer as a power of 10.



Australian fashion companies bring textiles, clothing and footwear into Australia from overseas. This is called importing. They also sell goods in overseas countries and this is called exporting. Both importing and exporting are controlled by laws and can involve paying taxes. It is illegal to import fake goods or copies of brand-name items.

Fashions may be cheaper from overseas

Where you see an 'idea' provided for a question, complete the calculations in it to find the answer.

- 1 a How many counterfeit items of clothing are brought into Australia each year?

Idea: $\sqrt{64\ 000\ 000\ 000\ 000}$

- b How much do illegal imports of clothing cost the local fashion industry each year?

Idea: \$ $\frac{10^2 \times 10^4 \times 10^6}{10^3}$



- 2 What is the difference in millions of dollars between the value of Australian imported goods and the value of Australian exported goods?

Idea: It is a number between 700 and 800 and is only divisible by 2 and 19.

- 3 It takes 10 minutes to make a T-shirt.

- a How much does it cost to make a T-shirt if a local worker is paid \$8 per hour?
b RIPOFF PTY LTD has moved production of its T-shirts to sweatshop workers overseas, who are paid 18 cents per hour. How much does it cost to make a T-shirt there?
c If such a T-shirt is sold for \$5, how much profit does RIPOFF make on each T-shirt?

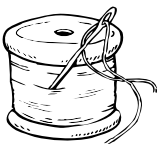
- 4 In 2005, sales of clothing, textiles and footwear in New Zealand totalled approximately \$2.5 billion (thousand million). If Australian products made up 19% of their imported clothing and 10% of their imported footwear, how much:

- a was the imported clothing worth?
b was the imported footwear worth?

- 5 BLING has been importing costume jewellery for many years. The table shows their latest order.

Number	Item	Cost per item (\$A)	Total cost
120	Necklaces and chokers	\$10.00	
55	Necklaces with earrings	\$12.00	
225	Crystal and glass necklaces with earrings	\$9.00	
Total			

- a Complete the table to find the total cost of the order.
b How much will the order cost altogether if BLING is charged \$2.29 per item for postage?



What about the taxes and duties?

- 6 The import duty on shoes is 10% of the cost. HIQUAL imported 500 pairs of shoes at \$43.93 per pair.
- a What import duty must HIQUAL pay altogether?
 - b What is the cost of a pair of shoes, including import duty?
 - c To make a profit of 200%, what should the advertised price of a pair of shoes be?

- 7 Trudy imported her bridesmaid's dress from the makers LUVLEE in Vietnam. LUVLEE charged Trudy \$55.80, including postage. She then had to pay 17.5% import duty. After that was added, she had to pay 10% GST (Goods and Services Tax). How much did her imported dress cost her in total?

- 8 BLING imported 350 pure silk sarongs and 120 vintage shawls from China. The sarongs were \$16 each and the duty to be paid was 10%. The shawls were two for \$60 and the duty was 17.5%. Find:

- a the total cost of the sarongs, including duty.
- b the total cost of the shawls, including duty.
- c the total cost of the order, including duty.



- 9 When goods arrive from overseas they are inspected by customs and a tax must be paid according to a percentage of the value of the goods. This tax is called a tariff or import duty. The table at the right shows the schedule of tariffs on different items of textiles, clothing and footwear.

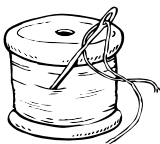
- a USTAR is importing shirts valued at \$15.35. How much tariff will be charged on each shirt?
- b USTAR decides that rather than importing the made-up shirt, they will import the fabric for \$6.53 per shirt and the sewing thread for \$0.48 per shirt. How much will the tariff be on each shirt then?
- c The tariff on clothing is reducing to 10% and all the other tariffs are reducing to 5% of the value of goods. USTAR needs 500 shirts.
 - i How much import duty will they pay for the made-up imported shirts?
 - ii How much import duty will they pay for the fabric and sewing thread for the shirts?
 - iii What is the difference in the import duty between importing made-up shirts and just the fabric and thread?

... FAST FASHION FACTS

A **tariff** or **import duty** is a government tax on goods being imported into a country. It is usually expressed as a percentage of the value of the imported goods and can be used to protect local industry.

Schedule of tariffs

Clothing	17.50%
Fabrics and cotton sheets	10.00%
Sleeping bags, table linen	7.50%
Carpet	10.00%
Footwear	10.00%
Footwear parts	7.50%
Other (e.g. yarns, thread, leather)	5.00%



Size charts usually give standard measurements for the actual circumference of the body rather than the size of the garment. Designers describe the space between your actual body measurements and a finished garment as ease. Ease is important in the design process as it results in the dimensions we see and how a garment behaves on the body. Ease is also essential for good body movement and comfort.

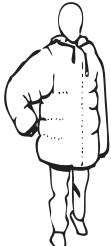
Tegan and Josh go shopping

The table below shows the space allowed between the actual body measurement and the finished clothing (ease) for different silhouettes. Because designers frequently use inches, the table of industry standards shows the ease as fractions of inches, but in this table the exact conversions to centimetres are also shown.

... FAST FASHION FACTS

- Ease is the space between the body measurement and the finished clothing. It varies between clothing styles and between designers.
- Stretch fabrics require little or no ease. This can also be true of knitwear.
- **Silhouettes** are the overall outside shape of the clothes. They can also be called **design lines**.
- The amount of material and hence the cost depends on the shape of the clothes.

Guideline Chart for Space Allowed for Ease

Description	Tight		Form fitted		Loose		Baggy		Oversized	
	Inches	cm	Inches	cm	Inches	cm	Inches	cm	Inches	cm
Men's Silhouettes										
Jackets	0–2	0–5.08	2 ¹ / ₈ –3 ¹ / ₂	5.40–8.89	3 ⁵ / ₈ –4 ¹ / ₂	9.21–11.43	4 ⁵ / ₈ –6 ⁷ / ₈	11.75–17.46	7 +	17.78 +
Pants	0–1 ¹ / ₂	0–3.81	1 ³ / ₄ –2 ¹ / ₂	4.45–6.35	2 ³ / ₄ –3 ¹ / ₂	6.99–8.89	3 ³ / ₄ –4 ¹ / ₂	9.53–11.43	4 ³ / ₄ +	12.07 +
Shirts	0–2	0–5.08	2 ¹ / ₄ –3 ¹ / ₂	5.72–8.89	3 ³ / ₄ –4 ¹ / ₂	9.53–11.43	4 ³ / ₄ –6 ¹ / ₂	12.07–16.51	6 ³ / ₄ +	17.15 +
Tops	0–2	0–5.08	2 ¹ / ₄ –4	5.72–10.16	4 ¹ / ₄ –5 ¹ / ₂	10.80–13.97	5 ³ / ₄ –7 ¹ / ₂	14.61–19.05	7 ³ / ₄ +	19.69 +
Knitwear	0–1	0–2.54	1 ¹ / ₄ –2 ¹ / ₄	3.18–5.72	2 ³ / ₄ –4	6.99–10.16	4 ¹ / ₄ –5 ¹ / ₂	10.80–13.97	5 ³ / ₄ +	14.61 +
Women's Silhouettes										
Jackets	0–1 ¹ / ₂	0–3.81	1 ³ / ₄ –2 ³ / ₄	4.45–6.99	3–4 ³ / ₄	7.62–12.07	5–6 ¹ / ₂	12.70–16.51	6 ³ / ₄ +	17.15 +
Pants	0–1 ¹ / ₂	0–3.81	1 ³ / ₄ –2 ¹ / ₂	4.45–6.35	2 ³ / ₄ –3 ¹ / ₂	6.99–8.89	3 ³ / ₄ –4 ¹ / ₂	9.53–11.43	4 ³ / ₄ +	12.07 +
Skirts	0–1 ¹ / ₂	0–3.81	1 ³ / ₄ –2 ¹ / ₂	4.45–6.35	2 ³ / ₄ –3 ¹ / ₂	6.99–8.89	3 ³ / ₄ –4 ¹ / ₂	9.53–11.43	4 ³ / ₄ +	12.07 +
Shirts/dresses	0–1 ³ / ₄	0–4.45	2–3 ¹ / ₂	5.08–8.89	3 ³ / ₄ –4 ¹ / ₂	9.53–11.43	4 ³ / ₄ –6 ¹ / ₂	12.07–16.51	6 ³ / ₄ +	17.15 +
Knitwear	0–1	0–2.54	1 ¹ / ₂ –2 ¹ / ₂	3.81–6.35	3–4	7.62–10.16	4 ¹ / ₄ –5 ³ / ₄	10.80–14.61	6 +	15.24 +



- 1 **a** What is the average (mean) ease in centimetres for a loose-fitting men's shirt?
b What is this measurement to the nearest inch?
- 2 **a** What is the average (mean) ease in centimetres for a baggy women's dress?
b What is this measurement to the nearest inch?
- 3 Tegan's size 10 evening shirt (bust 80 cm) is designed to be form fitted. What is the garment's average bust measurement after allowing the appropriate ease in centimetres?
- 4 Josh's skater jeans are worn loose and the designer uses the average ease. Josh measures 30 inches in the waist. What is the jeans waist measurement to the nearest inch?
- 5 Tegan's pencil skirt (size 10 for 60 cm waist and 80 cm bust) is worn tight, while her shirt is worn form fitted. Tegan's favourite designer develops fashions that are worn close to the body and always uses the average ease measurement for the skirt, but the lowest ease measurement for the shirt.
a What is the actual waist size of Tegan's skirt in centimetres?
b What is the actual bust size of Tegan's shirt in centimetres?
- 6 Tegan spots a funky jacket on a mannequin and it suits her beautifully. The jacket style is called Baby Doll; tight under the bust with a slight flare at the hip. Perfect! Tegan tries on a size 10. It is form fitted and has the full ease measurements applied. What would the jacket measure in the bust, in centimetres?
- 7 Josh has a 38 inch chest. Use the 'drop 6 principle' to find his waist measurement.
- 8 Josh's mate Rudy normally has a 36 inch chest. Rudy has been working out at the gym and now needs two more inches in the chest than the 'drop 6 principle' gives when applied to his 30 inch waist. What is Rudy's actual chest measurement in inches?
- 9 Josh likes a 'hip hop' jacket that is worn loose, made of faded denim with two back darts and patch pockets at the hip. The designer has used the lowest ease allowance. What is Josh's jacket measurement at the chest in inches?

... FAST FASHION FACTS

When developing male suits designers subtract **6 inches** from the chest size to get the waist measurement. This is referred to as the '**drop 6 principle**'.

Garments worn layered? Just enough space for perfume!

Designers consider the way garments will be coordinated, and so calculate the appropriate ease for each garment to ensure fit. Use the guideline chart on the previous page to answer the following.

- 10 The catwalk picture shows a woman's three-piece suit in a size 10 (60 cm waist and 80 cm bust). The jacket is a tight fit with the maximum ease, whereas the skirt is a form fit with a minimum ease. The skivvy is finely knitted with $\frac{1}{2}$ inch of ease. Use this information to calculate:
a the jacket's bust size with the relevant ease in inches.
b the skirt's waist size with the relevant ease in inches.
c the knitwear's bust size with the relevant ease in inches.





Ease is the space between the body measurement and the finished garment. A designer adds volume to certain types of garments to create the 'look' they seek, like in a bridal gown's full skirt. Ease is an important calculation in the making of fashion garments. It's all a matter of Maths!

body measurements + ease + designer's ease = actual garment measurements

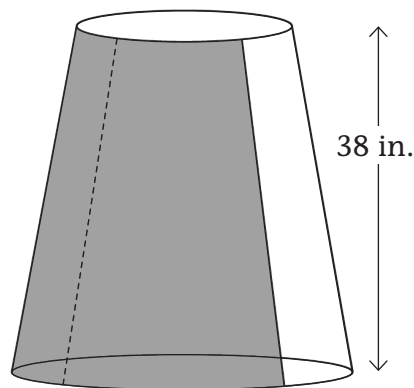
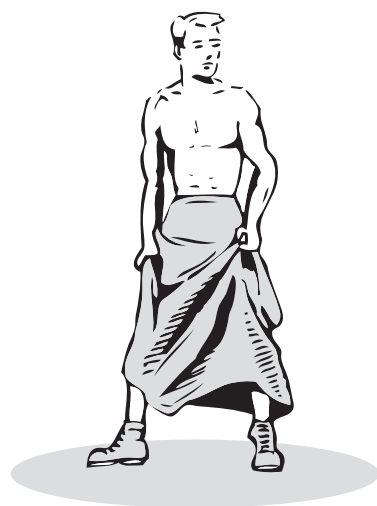
Tegan and Josh are off to Japan for the holiday season and will go shopping in Tokyo to see the famous fashion designs of JOJI-SUN. Walking past the windows of the chic department store PAKU in Tokyo Central, they see JOJI-SUN's awesome clothes. 'Okay let's try them on! You first!'

... FAST FASHION FACTS

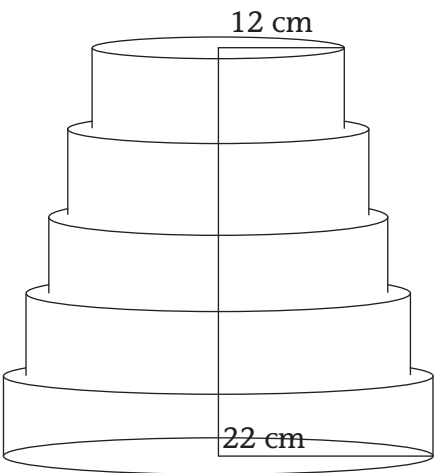
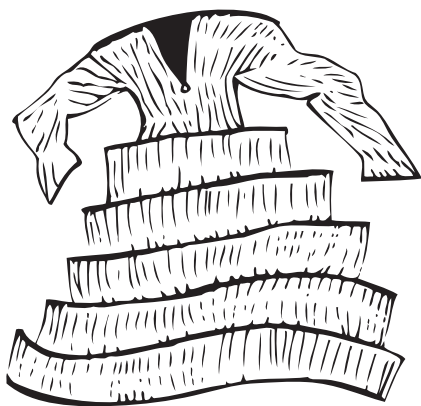
Ease is the space between the body measurement and the finished garment. Ease is used to make the garment fit tightly, be loose or even oversized.

Designer's Ease is the volume used to create the sculpture and drape we see in designs like those in the pictures on this page.

- 1 A JOJI-SUN dress is sized S, which is equivalent to an Australian size 8, with bust 75 cm and waist 55 cm. It is form fitted with an average ease measurement of 7 cm.
 - a What is the circumference of the dress at the waist including ease? Answer in whole centimetres.
 - b Hence use the formula $C = 2\pi r$ where C is the circumference and r is the radius to find the radius of the waist of the dress, correct to 2 decimal places.
 - c Calculate the dress's bust measurement, using the given ease.
 - d Hence find the radius of the dress at the bust in centimetres, correct to 2 decimal places.
- 2 The man's sarong style skirt that JOJI-SUN makes is shown in the picture. It is marked size M, which is equivalent to a man's 32 inch waist. The sarong is worn tight with an ease of $\frac{1}{2}$ inch and designer's ease of 1 inch.
 - a What is the total ease used in the sarong?
 - b Find the measurement of the actual waist of the sarong, including eases, correct to 1 decimal place.
 - c Hence find the radius of the actual waist of the sarong, including eases, correct to 1 decimal place.
 - d The sarong is flared so that the circumference of the hem at the bottom is 52 inches. Find the radius of the circle formed by the bottom hem.
 - e The drop from waist to hem of the sarong is 38 inches. There is an additional panel across the front which overlaps and measures 15 inches at the waist. Calculate the measurement of the actual waist of the sarong, including eases and the additional wrap panel, correct to 1 decimal place.

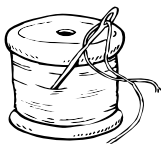


3 Another JOJI-SUN dress design has 5 garment panels from the waist to the floor, as in the picture below.



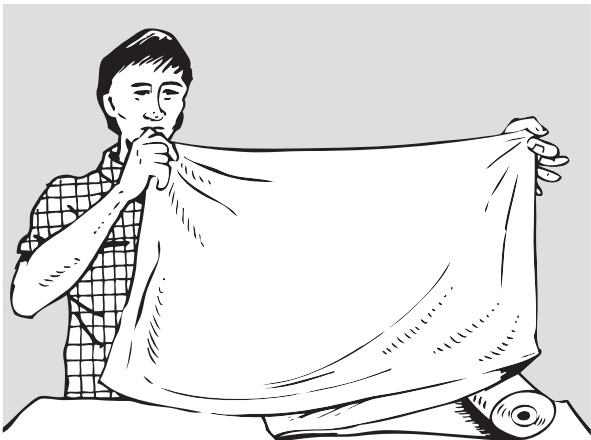
- a Each panel has a 9 cm drop. What is the total drop?
 - b The top panel has a radius of 12 cm. What is the circumference of the top panel, correct to 2 decimal places?
 - c The bottom panel has a radius of 22 cm. What is the circumference of the bottom panel, correct to 2 decimal places?
 - d By how much does the radius increase for each descending panel?
 - e By how much does the circumference increase for each descending panel, correct to 2 decimal places?
- 4 Calculate the circumference of the following panels to the nearest centimetre.
- a panel 2
 - b panel 3
 - c panel 4
- 5 Hence complete the table below to find the area of fabric in cm^2 used in the 5 panels.

Panel no.	Circumference (cm)	Area of fabric (cm^2)
1		
2		
3		
4		
5		
Total fabric area =		



Fashion designers do more than draw pictures (trade sketches) of the clothes they are creating. They need to make, or have made, samples for sale and promotion. This means that they need to understand what materials to use, how much they will need and how much extra they need to allow for sewing garments together. Other items like buttons and zips need to be arranged as well.

- 1 Dressmakers traditionally measure material by using the fact that 1 metre is approximately the distance from an adult's nose to the end of the fingers. Salim needed 2 metres of material to make a skirt. His nose-to-finger measurement is actually 103 cm.
 - a If he used the traditional method to measure the material, how much extra did he buy?
 - b How much did he pay for this amount of material if it cost \$17.50 per metre?

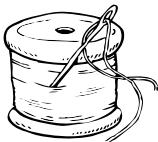


- 2 Dudette Designs has 5 new dresses designed for the coming season. The designers know that if they use 115 cm wide material, a dress needs 2 times the length of the dress plus 1½ times the length of the sleeve. If the dress length is 103 cm and the sleeve length is 52 cm, how many metres of 115 cm wide material is required to make:
 - a 1 dress?
 - b 5 dresses?
- 3 Each Skippi design skirt for the Japanese market uses 1.3 m of material. Because of the pattern on the material, 5% extra will be needed. How much material should be ordered altogether to make 500 skirts?

- 4 Dulcie makes alterations to dresses for a fashion shop. The price list is shown at the right. The shop recommends that people do not pay more than 5% of the value of a dress for alternations. Shiva really wants a \$350 dress with a multilayered skirt, but it is too long and the top side seam and shoulder seams need adjusting. Would the shop recommend that Shiva have the alterations done?

Price list	
Shorten skirt without lining:	\$15
Shorten skirt with lining:	\$25
Shorten multilayered skirt:	\$35
Lift skirt at waist:	\$35
Adjust top side seam:	\$15
Adjust top and bottom seam:	\$25
Adjust shoulder seam:	\$20

- 5 Material comes in different widths. For the shirt he is making, Lee has a choice of using 1.5 m of 150 cm wide material costing \$19.50 per metre or 2 m of 110 cm wide material costing \$14.50 per metre.
 - a Find the cost of the material for the shirt for each width of material.
 - b Find the area of material in m² Lee will receive if he chooses:
 - i 150 cm wide.
 - ii 110 cm wide.
 - c Find the cost per square metre for each width of material.
 - d Hence state which material is the cheaper choice.



- 6 Nicko has designed a new shirt called CHEMISE. He needs to work out how much it will cost to make. The material for the shirt costs \$32.00 per metre. It is a high quality cotton and the shirt requires 1.25 metres.
- a Calculate the total cost of material for the shirt.
 - b If 5% of this material will be wasted, what is the dollar value of the wasted material?
 - c Nicko chooses to sew the shirt together using the most expensive French seams. This involves 248 cm of sewing at a cost of 1.2 cents/metre. How much will the sewing of the shirt cost?
 - d How much will it cost for material and sewing for the CHEMISE shirt?

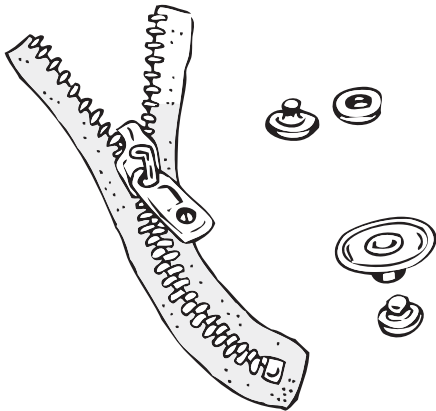
These jeans are going to look great with that shirt!

- 7 UDUDE Designs is developing a range of denim jeans to sell with their new shirt style. The design is called USTAR. The fit is baggy and worn low on the hip.
- a The denim from cotton mills in Tasmania costs \$7.50 per metre. USTAR needs 2.25 metres of denim for each pair of jeans. Calculate the fabric price for each pair of jeans.
 - b The jeans will have an Italian zipper by MIRRI, costing \$45.00 per 100, 4 medium-sized press studs at \$0.05 cents each and 4 large-sized press studs at \$0.10 cents each. For each pair of jeans, calculate the cost in dollars correct to 3 decimal places of:
 - i the zip.
 - ii the press studs.
 - c The cost of making the jeans is \$12.50 per pair, packaging is 50 cents per pair and freight and insurance is \$1.10 per pair. Complete the following table and hence find how much the USTAR jeans will cost UDUDE Designs for:
 - i one pair of jeans.
 - ii a run of 100 pairs of jeans.

... FAST FASHION FACTS

The word 'denim' came from the name of a French material, serge de Nimes, meaning serge (a type of material) from Nimes (a town in France).

	One pair (\$)	One run of 100 pairs (\$)
Cost of making jeans		
Cost of material	\$16.875	\$1 687.50
1 zip		
4 medium press studs	\$0.002	\$0.20
4 large press studs		
Packaging		
Freight/insurance		
Total cost of jeans		

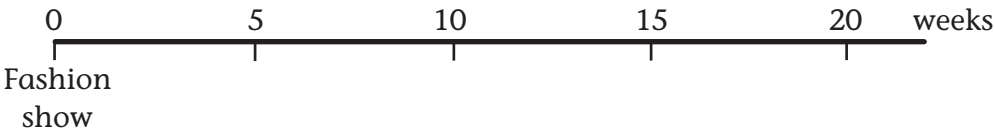


Everyone involved in the fashion industry, from fashion design studios to retailers, needs to cover their costs and to make a profit if they are to stay in business. Money is spent on design and making, mark-ups added by wholesalers and margins added by retailers. The companies involved could be in different countries and some of the processes could be completed by the same company – but still apply the margins and mark-ups! Decisions are made to ensure the best cost, quality and scheduling of production to match the market’s needs.

Fast fashion

- 1
- Worldwide, fashions usually change four times each year, usually according to the seasons. This means that taking a fashion range from the original fashion show to the consumer must happen quickly.
- a
- Show the information in the table on the timeline below.

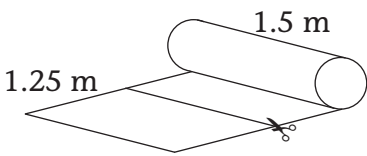
From	To	Time (weeks)
Fashion show	Design	0.5
Design	Factory	2
Factory	Shipping	5
Shipping	Shop	1
Shop	Factory outlet	9
	Total	17.5



- b
- If a fashion item remains in the factory outlet for only one week before being sold, what is the lifetime of the item from fashion show to purchase?

Taking the shirt off your back!

- 2
- UDUDE Designs is developing a collection of shirts for Australian males aged 15–19 years. The first style is called JACK. Each JACK shirt requires 1.25 metres of material from a 150 cm wide bolt (roll). The material costs \$5.00 per metre. Calculate the fabric cost for these shirts.
- 3
- The total cost of producing the JACK shirt is called the cost of goods sold (COGS). This cost will tell UDUDE how much it can charge the wholesaler for the shirts. They will use the local on-shore manufacturer called MAKE...
- a
- Complete the following table to calculate the COGS if the shirts are made on-shore by MAKE...



On-shore by MAKE...

CMT (cut, make and trim) a shirt	\$15.00
Material (fabric)	
Packaging (bought by designer or manufacturer)	\$0.50
Freight	\$0.60
COGS (cost of goods sold) is the total of these:	

... FAST FASHION FACTS

CMT means cut, make and trim. It often includes threads, tapes, some linings and some buttons unless otherwise specified by the designer.

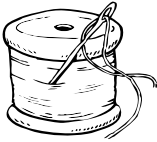
On-shore means products made within our country.

Off-shore means products made overseas.

COGS means cost of goods sold.

A wholesaler buys from the clothing maker and sells to the retailer.

A retailer sells to the public, usually in a shop.

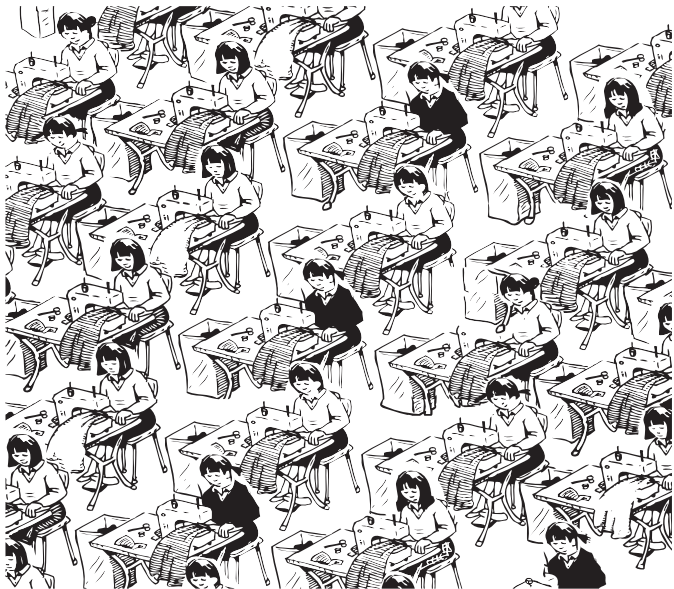


- b What is the total COGS for on-shore manufacturing?
- c What is the total COGS for UDUDE to make one run of 100 JACK shirts?
Answer in whole dollars.

Manufacturing off-shore

4 The JACK shirt will be made from material obtained from an Egyptian mill called PHAROS, which is 10% cheaper per metre than the on-shore costs given in Question 2. What is the price per shirt for fabric from PHAROS, correct to the nearest cent?

5 The shirts will be manufactured off-shore at a factory in Shanghai called SEWQUIK. The cost of cutting, making and trimming (CMT) will be 35% less than the on-shore costs (CMT in Question 3).

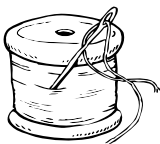


- a What is the off-shore manufacturing price for a shirt from SEWQUIK?
- b Find the total of the costs of cutting, making and trimming (CMT), the material, the packaging and the freight if packaging is 50 cents per shirt and the freight is 75 cents per shirt.
- c Since the shirts will be imported fully finished, duties and taxes must be paid and these will amount to 5% of the total found in Part b. Find the duties and taxes payable.
- d Hence complete the following table.

Off-shore by SEWQUIK

CMT (cut, make and trim) a shirt	
Material (fabric)	
Packaging	\$0.50
Freight	\$0.75
Duties and taxes (5% of the sum of all the above prices)	
COGS (cost of goods sold) is the total of these:	

- e What is the off-shore COGS price?
- f What is the total COGS price to UDUDE for one run of 100 JACK shirts when made in Shanghai? Answer in whole dollars.



In any one season, designers usually produce a range of clothing with a particular theme for a particular target group. For example, in this topic, UDUDE's target group is people aged 15–20 years and the theme is 'Cowboy Chic'. All the clothing in this collection will have cowboy-like details such as Texan emblems, checked patterns for shirts and Laredo boots.

The designer will also give the designs different names like 'rancher' or 'pancho' to give them a personality within the fashion range and to help sales assistants quickly identify designs in the shop. Designers must predict how many products are needed to form the range and in what colours, quantities and prices. This is an important part of their job.

- 1

Hi-Y's cropped pants in their PARTY range feature plenty of pockets, a tie belt and balloon pattern. Sizes are 3, 4, 5 and 6 and each pair is priced at \$22.99. How much will it cost to buy one pair of pants in each available size?
- 2

'Hi-Y has an exciting new fashion range hitting the stores this month,' said the headline. Its sports-inspired clothing has four pieces, each in two colours and in sizes 3, 4, 5, 6.

a

How many pieces of clothing make up the entire range?

b

If a third colour is added to the range, how many pieces of clothing will be in the range then?
- 3

Eleanor loves to wear the latest fashion from Hi-Y. Her mother has bought her 4 new skirts, 5 new shirts, 3 new pairs of shoes and 2 new pairs of socks for the coming season. If all the colours match, how many different combinations of outfits can Eleanor put together with these items of clothing?



UDUDE's Cowboy Chic range

UDUDE is developing a fashion range called 'Cowboy Chic' for the local market for summer. Their research has given the different numbers they will need in each size. The table shows the number of each size in the buying packs.

Men's size	28	30	32	34	36	38	Total no. in a buying pack
Quantity per pack	2	4	6	8	4	2	26
Women's size	8	10	12	14	16		Total no. in a buying pack
Quantity per pack	2	2	4	3	1		12

... FAST FASHION FACTS

Denim is actually **white** but is dyed indigo to create the inky blue colour of **blue jeans**.

Coloured jeans like red or white are usually made from a stiff 100% cotton to get the strong colour and soft feel needed.

Buying packs give the mixture of sizes of a particular style that need to be produced for a fashion season.

- 4

Complete the following statement:

A buying pack of men's clothing will contain ____ Size 30 and ____ Size 36.
- 5

What is UDUDE's most popular:

a

men's size?

b

women's size?
- 6

What whole number percentage of the men's buying pack will be Size 34?
- 7

What whole number percentage of the women's buying pack will be Size 14?

For the denim collection, UDUDE is planning two different jeans styles called USTAR and UROCK. They will be in different materials, colours and sizes as shown in the table.

Style	Style 1 - USTAR	Style 2 - UROCK
Retail selling price	\$150.00	\$175.00
Material	1. Denim	1. Denim
	2. Cotton drill	2. Cotton drill
Colour	1. Denim: – Indigo – Enzyme wash	1. Denim: – Indigo – Enzyme wash
	2. Cotton drill: – Black – Graphite – Silver grey	2. Cotton drill: – Black – Graphite – Silver grey
Waist sizes:		
Men's (M) inches	28, 30, 32, 34, 36, 38	28, 30, 32, 34, 36, 38
Women's (W)	8, 10, 12, 14, 16	8, 10, 12, 14, 16



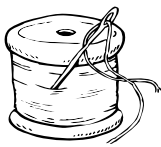
8 To get an idea of how much money they will make, UDUDE uses USTAR in indigo coloured denim for its calculations, because they expect it to be their most popular line. They need 3 buying packs of these. Use the information in the table on the previous page for 1 buying pack to fill in the following table for 3 buying packs.

USTAR jeans – selling price \$150

Men's size	28	30	32	34	36	38	Total
Number required for 3 buying packs							
Women's size	8	10	12	14	16		Total
Number required for 3 buying packs							

- 9 Hence, find the total number of pairs of USTAR jeans needed altogether for:
- a men. b women.
- c both men and women.
- 10 How much money will UDUDE receive from sales for all the jeans in the three buying packs of both men's jeans and women's jeans? Answer in whole dollars.
- 11 How much money will UDUDE receive by selling all the jeans in the three buying packs of only the most popular size for:
- a men? b women? c both men and women?
- 12 What percentage of all the money received from sales, which you calculated in question 10, is made by selling just the popular size for both men and women?





Designers usually produce a range of clothing in materials and colours that will be popular and will look good on their customers. Some colours can make a person look bigger and others make a person look smaller. A bright yellow object usually looks bigger than the same sized object in black! Predicting the correct mix in the range is an important part of the designer's job. In this section, UDUDE extends its 'Cowboy Chic' range for its target group of people aged 15–20 years.

- 1 CUTE Designs wish to use 4 shades of blue as the main or core colours in their collection, highlighted by 4 accent colours and white. How many combinations of 2 colours can they use if one must be a core colour and one must be an accent colour or white?



... FAST FASHION FACTS

- In a fashion range, colours are often divided into two main categories:
- **Core colours** are common colours across the collection. In the table below, **black**, **charcoal** and **grey** are core to the collection.
 - **Accent colours** are seasonal colours which help to highlight the core colours. In the table below, **taupe**, **pink** and **white** are accent colours to the collection.

UDUDE is developing a fashion range for a local summer season. The theme is 'Cowboy Chic'. As well as jeans styles, UDUDE will create two styles of shirts called RANGER and SANDY. These will be in different materials, colours and sizes as shown in the table.

- 2 A men's size S usually fits a 36–38 inch chest, but often a pattern maker will size the garment to fit a median (middle) chest of 37 inches. Use this fact to convert S (36–38), M (38–40), L (40–42) and XL (42–44) into the median chest inch sizes.
- 3 Use the fact that there are approximately 2.54 cm in 1 inch to convert your answers from question 2 into metric sizes to the nearest centimetre.

Style	Style 1 - RANGER	Style 2 - SANDY
Retail selling price	\$79.00	\$85.00
Material	100% cotton poplin	90% cotton + 10% lycra
Colour	Men's	Women's
	Black	Black
	Charcoal	Charcoal
	Taupe	Taupe
	White	White
Size Men's (M)	S, M, L, XL	Not available
Size Women's (W)	Not available	8, 10, 12, 14, 16

The following table shows the number of each size that will be put in a men's buying pack.

Men's size	28	30	32	34	36	38	Total no. in a buying pack
Number required per pack	2	4	6	8	4	2	26

... FAST FASHION FACTS

Buying packs give the mixture of sizes of a particular style that need to be produced for a fashion season.

4 In fact, for the style RANGER, the buying pack will need to contain 3 buying packs of the core colours and 2 buying packs of the accent colours. Using this information, fill in the following table for the different colours and sizes.

Style 1 - RANGER Shirts - Selling Price \$79

Men's size	28	30	32	34	36	38	Total
No. required per buying pack	2	4	6	8	4	2	26
No. required for 3 buying packs Poplin – black (core)	6	12	18	24	12	6	78
No. required for 3 buying packs Poplin – charcoal (core)							
No. required for 2 buying packs Poplin – taupe (accent)	4	8	12	16	8	4	52
No. required for 2 buying packs Poplin – white (accent)							
Totals							
No. as a percentage of the total number							
Net sales return = number × RSP							

- 5 Hence read off:
- a the total number of men's shirts for each size.
 - b the total number of shirts altogether for the entire style RANGER.
- 6 By multiplying the total number of men's shirts by the selling price of \$79, find how much money UDUDE will receive from sales of each size.
- 7 Hence find the total amount of money UDUDE will receive from sales for the entire style RANGER.
- 8 Hence find the total amount of money UDUDE will receive from sales for the entire style SANDY and write the values into the table.

Style - SANDY Shirts RRP \$85

Women's size	8	10	12	14	16	Total
Total number of each size	20	24	40	34	6	124
Amount received on sales = number × selling price						

- 9 Use your answer to Question 8 to calculate the total money received from sales of the two styles, RANGER and SANDY, altogether.
- 10 What percentage contribution of UDUDE's total sales of shirts is made from SANDY? Round your answer to the nearest whole number.



Most material we use is woven and is sold in lengths (and widths). Woven material will fray at the edges if it is not finished. Jumpers, sweaters and knitwear will all keep you warm, but unlike woven fabrics, yarns are sold by weight. Knitwear is expensive to produce since it is slow to knit and the raw materials are expensive.

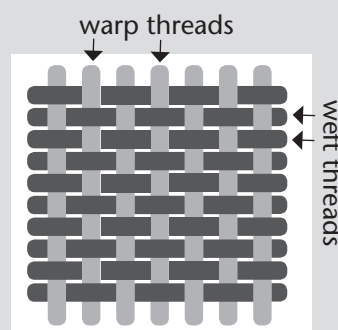
Warp and weft - it's all in the weave!

Thread count is measured by counting the number of threads contained in 100 cm^2 , including both the warp threads and the weft threads.

- 1 If the fabric pictured in the fast fashion fact has an area of 100 cm^2 , find the thread count of the fabric.
- 2 In the sentence below, is the missing word **coarser** or **finer**?
The higher the thread count, the _____ the threads.
- 3 A 100 cm^2 piece of fabric has a thread count of 250 and weighs 0.5 g. How much would 1 m^2 of the same fabric weigh?
- 4 Cleopatra's bed sheet has a thread count of 250 and measures 2.5 m wide and 3.5 m long. Find:
 - a the area of her bed sheet in m^2 .
 - b the weight of her bed sheet in grams if it is made of the same fabric as in question 3.

... FAST FASHION FACTS

In woven materials, the threads that run the length are called the **warp** and those that run edge to edge are called the **weft** (or **woof**).



North is warp, left is weft!

Are you pulling the yarn over my eyes?

- 5 Beauticraft Yarns sells balls of 98% acrylic, 2% polyester yarn in 170 g balls. Each ball contains 169 m of yarn.
 - a Approximately how much will 100 m of the yarn weigh?
 - b When the yarn is knitted correctly on 6 mm needles, 20 rows of knitting will measure 10 cm. How many rows will be needed to knit a sleeve that is 42 cm long?
- 6 You will need the following quantities of 75% acrylic, 25% wool yarn for knitting a boy's size 10 hoodie:
 - Main colour, black, 6 balls
 - Contrast colour A, grey, 1 ball
 - Contrast colour B, blue, 1 ball
 - Contrast colour C, red, 1 ball

Each ball weighs 100 g and costs \$7.95. You will also need 2 pairs of knitting needles costing \$6.98 each and a zipper costing \$11.00. If all the wool is used, how much will it cost to knit and make the hoodie?

... FAST FASHION FACTS

Yarn is made of strands twisted together, ready to be woven or knitted. It can be wool, mohair, cotton, linen, ramie, acrylic, nylon, polyester, etc. It is sold by the kilogram.

A **spool** is a conical-shaped object that **yarn** is spun around.





A French knitwear designer Fifi LaPearl produces a sample women's jacket that is 70's inspired, has a multicoloured collar and is fitted to the body. The style is called Madame X. The actual sample weighs 360 grams. There was 7% yarn wastage in the production method.

- 7 For 1 kg of yarn:
 - a what will be the total wastage in grams?
 - b how much usable yarn is available?
 - c how many complete pieces of Madam X knitwear can be made?
 - d how many grams of usable yarn will be left over?

The jacket is 'fully-fashioned', aimed at women at the upper end of the market and with a retail price of \$270.00. 50 units of Madam X knitwear have been ordered by a French department store called TRICOT.

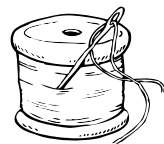
... FAST FASHION FACTS

Fully-fashioned is when knitwear pieces are individually made then sewn together. This is expensive to produce with a high % of wastage.

Cut-and-sew is when the yarn is knitted in one long piece and then cut into the required shapes to be sewn together. This is less expensive to produce with a relatively low % of wastage.

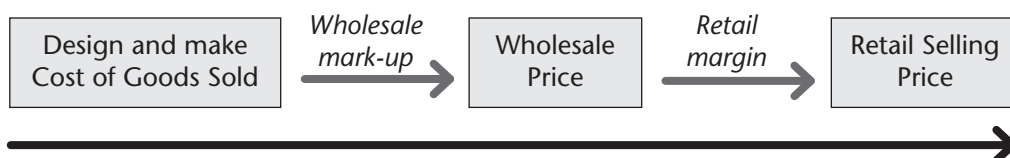
- 8 a What is the exact weight of this order, including wastage?
 - b How many 1 kg spools of yarn need to be bought to cover the order?
 - c The yarn is 90% wool, 5% cashmere and 5% lycra. It is made in Italy at a cost of \$22.50 per 1 kg spool. What will the cost of the yarn be?
 - d If TRICOT sells the jackets at \$270 each, how much money will they receive from the sales of the order?
- 9 The cost of the goods sold (COGS) includes the cost of labour, materials and freight. For fully-fashioned knitwear, the cost of goods sold is 35% of the selling price. This leaves 65% for the 'return', or the difference between the cost of goods sold and what it sells for.
 - a What is the COGS, given the selling price is \$270.00?
 - b What is the return per jacket for the selling price of \$270.00?
 - c What will be the return on the whole order of 50 jackets?
 - 10 If the same knitwear is made by the cut-and-sew method, it costs only 27% of the selling price to produce. For the cut-and-sew method, find:
 - a the COGS, given the selling price is still \$270.00.
 - b the new return per jacket for the selling price of \$270.00.
 - c the new return on the whole order of 50 jackets.





The retail selling price of a piece of clothing in a shop is the price you see printed on the swing tag or label. This price reflects the cost of designing, making, selling and distributing the item. Of course, the shop owner adds a retail margin to cover costs such as rent and wages, as well as their profit.

In this topic, you will see who makes the money and how much and how a shop adjusts its margins across a range of clothes to make the most amount of money! The fashion buyer needs to make many of these calculations as a part of the job. They are particularly interested in the return on the money invested in designing and making the goods. This is called ROI (return on investment).



Buying chain chart

- 1 A shirt which cost \$11.50 to design had a mark-up of 100% of the cost of goods.
 - a What was the wholesale price?
 - b If the shirt then had a retail margin of 100% of the wholesale price, what was the retail selling price?
 - c What was the ratio of cost of goods to retail selling price?

The retail selling price of a pair of trousers was \$153.00. The 'one-in-three principle' was followed. Use the fast fashion fact and the buying chain chart to answer the following questions.

- 2 What was the cost of goods price (COGS) for the trousers?
- 3 The wholesale mark-up and the retail margin were the same. What was:
 - a the mark-up?
 - b the margin?
- 4 What was the wholesale price?
- 5 What fraction of the retail selling price was the margin?
- 6 What was the percentage increase from the COGS to the retail selling price?
- 7 Find the difference between the cost of goods price and the retail selling price – that is, the return.
- 8 What fraction of the retail selling price is the return?

... FAST FASHION FACTS

The 'cost of goods sold' is often shortened to **COGS**.

The **one-in-three principle** means that the cost of goods sold is $\frac{1}{3}$ of the **retail selling price**. This is standard practice in the fashion business.

The one-in-three principle means that the **return** is $\frac{2}{3}$ of the retail selling price.

... FAST FASHION FACTS

Most fashion buyers are interested in the **return** on the money spent to get the clothing from the designer to the shop. They calculate the return as a percentage of the selling price:

$$\frac{\text{Retail Selling Price} - \text{COGS Price}}{\text{Retail Selling Price}} \times 100 = \text{Percentage Return (\%)}$$



9 Complete the following calculations correct to 2 decimal places.

a Retail selling price = \$28.50

Cost of goods price = \$9.50

Return = \$ _____

c Retail selling price = \$45.00

Cost of goods price = \$14.00

Percentage return = _____

= _____ %

10 The selling price of a shirt is \$80.00 and the cost price is \$25.00. What is the percentage return at the shop, correct to 2 decimal places?

11 The selling price of a pair of jeans is \$150.00 and the cost of goods price is \$38.00. What is the percentage return to retail, correct to 2 decimal places?

12 Fifi the fashion buyer will only deal with companies where the return is less than 70%. What is the highest cost of goods price she will accept for a T-shirt which will be sold at \$54?

b Retail selling price = \$30.00

Cost of goods price = \$10.00

Return as a percentage of the selling price

$$= \frac{30 - 10}{30} \times 100$$

= _____ %



Drop everything - ITECHU is having a sale!

13 ITECHU's puffer jacket with hood is its most expensive product, selling for \$250. At the sale it was marked down to \$162.50. Find:

a the markdown in dollars.

b the markdown as a percentage of the retail selling price.

c the sales revenue from the remaining 24 jackets.

... FAST FASHION FACTS

Reducing the selling price of a product is called a **markdown**.

Markdowns are used when certain items are not selling as quickly as expected or when other shops are selling similar products more cheaply and retail sales have slowed.

$$\text{Percentage Markdown (\%)} = \frac{\text{Amount of Markdown}}{\text{Retail Selling Price}} \times 100$$

Sales revenue is the total amount of money received if all the items are sold.

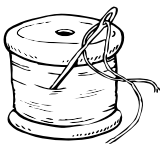
14 a ITECHU's Jean style #1 is being marked down by 30%. If the retail price is \$120, by how much (in dollars) has this style been marked down?

b What is the new retail selling price for this style?

c There are 87 pairs of Jean style #1 left.

i What is the total value of the markdowns for this style?

ii What is the sales revenue for these 87 pairs of jeans?



If a company is to make a profit, it has to make more money from selling its goods than it spends. If it spends more than it earns, then it will make a loss and could eventually go out of business. If there is a profit then it can be spent on new products, buying more stock, expanding the business, research and development or just put it in the bank.

In this topic, UDUDE is setting up a table to help keep track of their sales and expenses to make sure they continue to make a profit.

UDUDE is getting serious!

- 1 UDUDE is selling a new line called USTAR jeans. The cost of the goods that UDUDE sells is always one third of the price they sell them for. This is called the ‘one-in-three principle’ and is the usual formula used by fashion companies.

 - a If the cost of goods was \$10 000, how much will UDUDE receive from selling the goods?
 - b How many pairs of USTAR jeans were sold if they retail for \$150.00 each?
 - c Use the fast fashion facts to find the return to UDUDE on the \$10 000 spent.
 - d If UDUDE spends \$6 500 on costs that are fixed like equipment and rent, and \$11 000 on costs that can be varied like staff salaries and advertising, how much does it cost them to operate their business (operating expenses)?
 - e What percentage of sales are the operating expenses?
 - f If the profit before tax is the return – operating expenses, how much profit did UDUDE make on these sales?
 - g UDUDE will use a table to help keep track of their sales and spending. This is called a ‘profit and loss statement’. Use your answers to Parts a to c to fill in the table.

... FAST FASHION FACTS

The ‘cost of goods sold’ is often shortened to **COGS**.

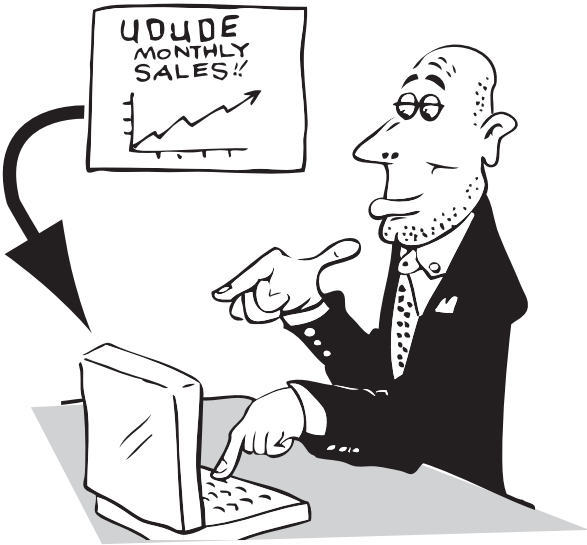
The **one-in-three principle** means that the cost of goods sold is 1/3 of the **retail selling price**.

The **return on sales** = amount received from sales – the cost of goods sold.

Profit before tax = return on sales – operating expenses

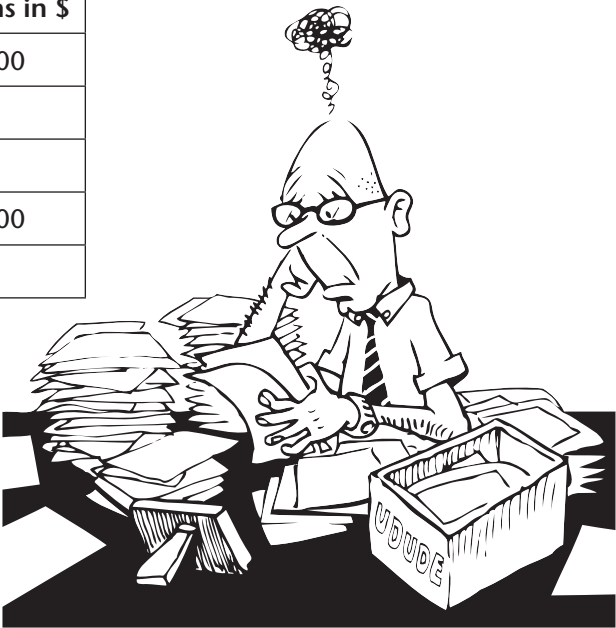
UDUDE’s first profit and loss statement

	Calculations in \$
Amount received from sales	30 000.00
Cost of goods sold (COGS)	
Return = Sales – COGS	
Operating expenses	17 500.00
Profit before tax	



h The cost of goods is reduced by 15%. Complete the statement below to show the new return and new profit.

	Calculations in \$
Amount received from sales	30 000.00
Cost of goods sold (COGS)	
Return = Sales – COGS	
Operating expenses	17 500.00
Profit before tax	



What 7%? I'll go broke!

The cost of goods is still \$10 000 and UDUDE still receives \$30 000 from selling the goods. But they want to make a profit before tax of 10%.

- 2 Complete the following to find how much profit they need to make.

Profit (\$) = 10% of amount received from sales
 = $\frac{1}{10} \times \$$ _____
 = \$ _____
- 3 To give this profit, what would the operating expenses need to be, in dollars?
- 4 The government charges company tax at 30% of the profit before tax.

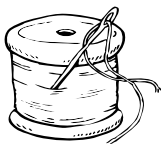
a How much company tax is due?
b What is UDUDE's net profit in dollars after that?
c What is UDUDE's net profit **after tax** in dollars, as a percentage of amount received from sales?
- 5 Complete the table below.

UDUDE's second profit and loss statement

	Calculations in \$
Amount received from sales	30 000.00
Cost of goods sold (COGS)	10 000.00
Return = Sales – COGS	20 000.00
Operating expenses	
Profit before tax	
Profit after 30% company tax	

... FAST FASHION FACTS

Net profit = Cash to splash!
In Australia, **company tax** is usually 30% of the profit before tax.



UDUDE will open its first retail store in Chapel Street Prahran, 3181. In order to develop an image that will last in the market place, the UDUDE business will need to consider a range of factors from fashion design to managing profits. We will see how UDUDE set up a fashion store and stock it with the sorts of clothes and accessories that will make them retail stars!

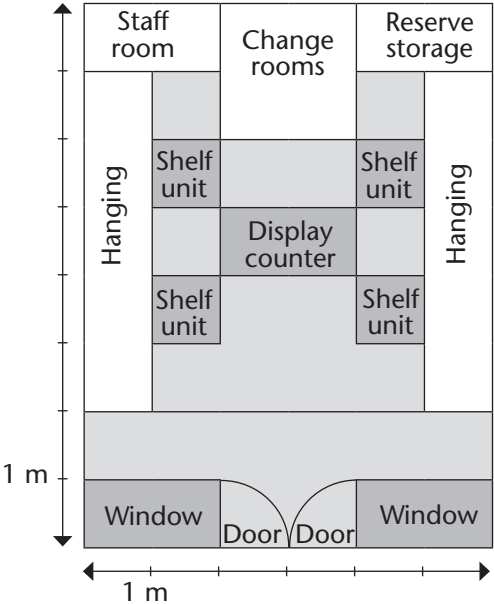
How much space have I got?

1 Chapel Street is the place! The retail space has a central door with deep windows for good display and great passing foot traffic. UDUDE-3181 will rent the shop with a business lease. The retail space for UDUDE-3181 is rectangular in shape. It runs 6 metres wide and 8 metres long. How many square metres is the store?

2 The plan of the shop before the merchandise has been bought is as shown in the scale diagram at the right.

- The staff room occupies 2 m².
- The change room occupies 4 m².
- The reserve for stock occupies 2 m².
- The counter occupies 2 m².
- Each hanging space occupies 5 m².
- Each shelf unit occupies 1 m².

- a Calculate the total area taken up by the spaces listed above.
- b Each door swings through an angle of 90° and the width of each door is 1 metre. Use the fact that the area, $A\text{ m}^2$, of a circle of radius $r\text{ m}$ is given by $A = \pi r^2$ to calculate the floor space that the swing of two doors takes up.
- c Hence use subtraction to find the overall area available for selling. Express this as a percentage of the total area of the shop.
- d UDUDE-3181 expects to make \$520 000. Use division to find out how much money each square metre of selling floor space will make.



UDUDE is ready to stock the shop. Here come the bills!

3 The interior designers have completed their work and created two long hanging walls with 5 metres of hanging space on each. They have also created 4 free-standing fixtures (base area 1 m²). These will allow the merchandise to be displayed in shelves and stored.

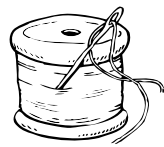
- a How many usable linear metres of hanging space do UDUDE have in location 3181?
- b UDUDE-3181's assortment of stock is shown at the right. The jackets, pants, half of the jeans and half of the shirts will be hung up. What percentage of the collection will be hung?
- c To find out what this assortment plan means in dollars for UDUDE, we need to use the sales expected for the year of \$520 000. Use this figure and the percentages in the assortment to calculate how much money will be received in each of the seven categories.

... FAST FASHION FACTS

An **assortment plan** is the stock making up the collection, expressed as a percentage of the whole.

Assortment of stock

- 1. Jackets – 30%
- 2. Pants – 15%
- 3. Jeans – 15%
- 4. Shirts – 20%
- 5. Knitwear – 10%
- 6. T-shirts – 5%
- 7. Accessories – 5%



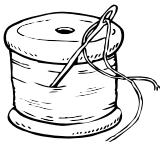
3181 - it's all in the number! The fashion postcode!

4 UDUDE has carried out its market research and found that there is good demand for their fashions in the Prahran area. They have had to calculate how much they are likely to return in sales, and what their operating costs will be. The following table shows their estimations for one year. Complete the table to calculate how much money they expect to receive on average from sales:

- a each week. b each month.

Estimated business facts	Dollar value	Percentage of income
Income from 1 year of sales at UDUDE's new store in 3181	\$520 000.00	
Occupancy costs for 1 year – renting and using the shop leads to costs such as rent and bills like rates, gas, electricity, water and sewerage.		5.000%
Fixed and variable costs – for staff, advertising, packaging, fixtures, insurance, etc. These are the costs of running the business.	-\$242 840.00	
Cost of goods sold (COGS) – cut, make and trim, labour, clothes, materials and freight.		33.333%
The return is the money returned after the cost of goods is subtracted from the money received from sales.		66.667%
Gross profit is the money left after all the expenses (fixed, variable and occupancy) have been subtracted from the return.		
Company tax is 30% of gross profit.		
Net profit – 'cash to splash' – money left after all the taxes are paid!		

- 5 Because of the demand for UDUDE's clothes, they believe that the projected annual sales are reasonable. They now need to calculate their yearly expenses.
- a UDUDE's sales for the year are estimated at \$520 000.00. The occupancy costs for the shop can be estimated at 5% of that figure. How much can UDUDE expect to pay in occupancy costs for one year?
 - b Calculate the total cost of goods if the one-in-three principle states that it should be 1/3 of money received from sales.
 - c Calculate the return by subtracting the COGS from the money received from sales. This should be 2/3 of the money received from sales.
 - d Fill in the blanks in the table. Hence calculate the gross profit for UDUDE's sales in the first year.
 - e What is the gross profit as a percentage of the first year's sales, correct to the nearest whole number?
 - f Assuming company tax of 30% of gross profit, calculate this tax for the year in dollars.
 - g Hence find the value of the net profit and write it in the table.
- 6 UDUDE-3181 will stay in business if they make a net profit of 10 cents in the dollar of sales. This means that the net profit should be 10% of the money received in sales.
- a What is the net profit as a percentage of sales?
 - b Is this profit enough for UDUDE-3181 to stay in business?



1 - Funny fashion facts

- 1 18th 2 1916 3 2000 4 1880 5 6 6 16
- 7 1785 8 2.724 9 5-9 metres 10 1933 11 1929
- 12 a 3106 carats b 621.2 grams
- 13 10 14 \$10 million
- 15 Rayon (1910), Nylon (1939), olefin (1949), acrylic (1950), polyester (1952) spandex (1958) and aramid (1961)
- 16 1794 17 4.00% to 5.00%
- 18 No; natural shrinkage to 79.54 cm
- 19 a 2.40 m² b 1.14 kg 20 \$2222

2 - Maths you need in the industry

- 1 a 10.16 cm b 10.2 cm c 10.1 cm
- 2 a 101.6 cm b 102 cm c 101 cm
- 3

Size	Inches	Centimetres
7	9.875	25.0825
8	10.125	25.7175
9	10.5	26.67

- 4 Conversion factor accurate to more decimal places because of no interpolation (guessing between) needed; multiplying magnifies errors
- 5 \$200.65 6 No; the price will be \$87.50
- 7 a 31.25% b 2.5% c 20 size 12s and 15 size 14s
- 8 1.302 m 9 \$30 900 10 \$38.25 11 \$31.50
- 12 a \$32 b 20% c \$208
- 13 a \$78 b \$80.34 c \$83.96

3 - Fast fashion statistics

- 1 \$457.38 2 88.2%
- 3 a (i) 4250 (ii) 4250 (iii) 2550 b 5950
- 4 a 194 444 444 b \$330 billion
- 5 2666 6 2500 7 0.012 or 1.2% 8 \$31 818.18
- 9 \$200 000 10 22 000 - 22 400
- 11 more than 18 400 12 10 000 13 \$21.89
- 14 (from left to right) \$30 000; \$42 140; \$62 640; \$87 510; \$117 120
- 15 a less than \$42 140 b more than \$87 510 c less than \$117 120 d more than \$30 000

4 - Stepping into the fashion world

- 1 a NSW b SA 2 a graphic design b 6244
- 3 TAS 4 ACT 5 a 12
- b

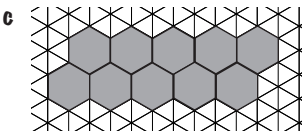
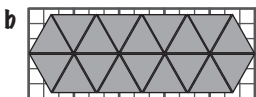
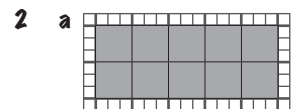
cm	22.8	23.1	23.5	23.8	24.1	24.5	24.8	25.1	25.4
US	5	5½	6	6½	7	7½	8	8½	9

cm	23.5	23.8	24.1	24.5	24.8	25.1	25.4	25.7	26.0
UK	3½	4	4½	5	5½	6	6½	7	7½

- d 9 ½
- 6 a \$14,246.00 b \$26,115.00 c 35.3% d 64.7% e \$2475.00

5 - Traditional designs - Maori, Mayan & Amish

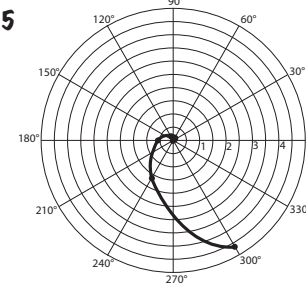
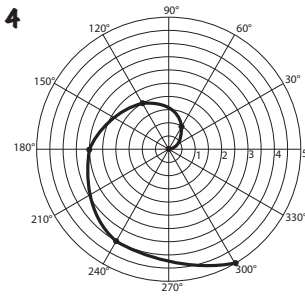
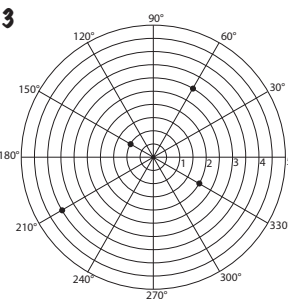
- 1 a No answer given b 3800 cm² c 7400 cm² d 180 cm



- 3 a AB and GC b 2 4 7.5 cm
- 5 a 120 b 480
- 6 81 000 cm² 7 no answer given

6 - Traditional designs - Navajo & Hmong

- 1 a \$18 640 b 43 cm square woven picture at 29.7 cents per square cm
- 2 a \$200 -\$255 b less than \$28
- 3



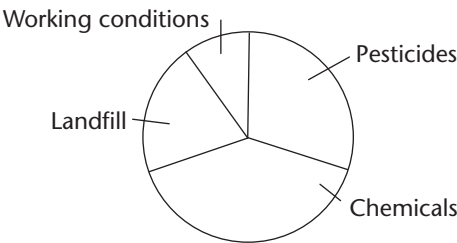
7 - Future fashion

- 1 between \$42 million to \$87 million
- 2 a 73.42 b 33.33 kg
- 3 a 1440 b chemical used to bleach, colour and print

4.5

Effect	Number	Fraction	Degrees
Pesticides	432	0.3	108
Chemicals	576	0.4	144
Landfill	288	0.2	72
Working conditions	144	0.1	36
Total surveyed	1440	1	360

- 6 Number of students who felt this was the most important effect



- 7 a 2.54 m² b 431.8 g
- 8 a 142 350 km b 4 c 500 mL d (i) 4.5 mL (ii) 11.4 mL e No
- 9 a 1/3 nm b 300 000 c 1178 nm³



8 - What is my size?

1 a

Standard size	6	8	10	12	14
to fit Bust (cm)	70	75	80	85	90
to fit Waist (cm)	50	55	60	65	70
to fit Hip (cm)	75	80	85	90	95
European size	36	38	40	42	44

b 60 cm c 33 inches d Size 10 e 31½, 23½, 33½
f 85 cm g 82.5 cm

2 a

Size	6	7	8	9	10	11	12	13	14
Waist (cm)	50	52.5	55	57.5	60	62.5	65	67.5	70

b 87.5 cm c 34½ inches

3 a

Chest	Imperial (inches)		36	38	40	42	44
	Metric (cm)		92	97	102	107	112
	European sizes		46	48	50	52	54
	Exact Values (cm)		91.44	96.52	101.60	106.68	111.76
Waist	Imperial (inches)	28	30	32	34	36	38
	Metric (cm)	72	77	82	87	92	97
	Exact Values (cm)	71.12	76.20	81.28	86.36	91.44	96.52

b 32 inches c 107 cm d 92 cm e 46

9 - Body measurements

- 1 175.3 cm 2 27.8 cm 3 2.9 cm 4 46.3 cm
5 13 cm 6 9.3 cm 7 74.1 cm 8 68.5 cm
9 43 cm
10 a 44" (42.95") b 38" to be sure (actual answer 36.02")

11, 12

Men's Body Measurements

Chest	34"-36"	38"-40"	42"-44"	46"-48"	50"-52"
Waist	28"-30"	32"-34"	36"-38"	40"-42"	44"-46"
Neck	14½-15"	15½-16"	16½-17"	17½-18"	18½-19"
Sleeve	32"-33"	33"-34"	34"-35"	35"-36"	36"-37"

- 13 No; too long
14 No; need to be altered
15 30 inches
16 a 34 inches b 40 inches

10 - Getting the colour right

- 1 18 years old in 1856 2 275
3 a 25 b 1500
4 a \$23,760 b \$7920
5 Yes; would have saved \$3240
6 a 7500 b \$168 750 c \$166 250
7 900
8 a 16,777,216 b black (no red, no green, no blue) c white
9 a Red-brown, Lolly Pink and Golden Yellow
b (249,94,89), (237,114,170), (255,237,56) c 10¹²

11 - Importing and exporting

- 1 a 8 million b \$1 billion (1000 million)
2 722 million
3 a \$1.33 b \$0.03 c \$4.97
4 a \$475 million b \$250 million
5 a

Number	Item	Cost per item \$A	Total Cost
120	Necklaces and chokers	\$10.00	\$1 200.00
55	Necklaces with earrings	\$12.00	\$660.00
225	Crystal and glass necklaces with earrings	\$9.00	\$2 025.00
		Total	\$3 885.00

- b \$4801
6 a \$2196.50 b \$48.32 c \$144.97
7 \$72.12
8 a \$6160.00 b \$4230.00 c \$10 390.00
9 a \$2.69 b \$0.68 c (i) \$767.50 (ii) \$175.25 (iii) \$592.25

12 - Design lines and silhouettes

- 1 a 10.48 cm b 4.12 inches
2 a 14.29 cm b 6 inches
3 86.99 cm
4 a 33⅛ inches
5 a 61.91 cm b 85.08 cm
6 86.99 cm 7 32 in. 8 38 in. 9 41⅝ in.
10 a 33 inches b 25 inches c 32 inches

13 - Design and sculpture

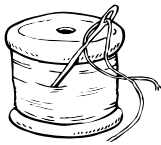
- 1 a 62 cm b 9.87 cm c 82 cm d 13.05 cm
2 a 1½ inches b 33½ inches c 5.1 inches d 8.28 inches e 48 ½ inches
3 a 45cm b 75.40 cm c 138.23 cm d 2.5 cm e 15.71 cm
4 a 91 cm b 107 cm c 123 cm
5 4812 cm²

14 - Seams too quick

- 1 a 6 cm b \$36.05
2 a 2.84 m b 14.2 m
3 682.5 m 4 No, alterations cost \$70 or 20%
5 a \$29.25; \$29.00 b (i) 2.25 m² (ii) 2.2 m² c \$13.00; \$12.89 d 110 cm wide
6 a \$40.00 b \$2.00 c \$2.98 d \$42.98
7 a \$12.50 b (i) \$0.45 (ii) \$0.002; \$0.004
c

	One pair	One run of 100 pairs
Cost of making jeans	\$12.500	\$1 250.00
Cost of material	\$16.875	\$1 687.50
1 zip	\$0.450	\$45.00
4 medium press studs	\$0.002	\$0.20
4 large press studs	\$0.004	\$0.40
Packaging	\$0.500	\$50.00
Freight/Insurance	\$1.100	\$110.00
Total cost of jeans	\$31.431	\$3 143.10

(i) \$31.431 (ii) \$3143.10



15 - Making clothes is a global experience

- 1 a Time line labels: design: 0.5; factory: 2.5; shipping: 7.5; shop: 8.5; factory outlet: 17.5
b 18.5
- 2 \$6.25
- 3 a No answer given b \$22.35 c \$2235.00
- 4 a \$5.63
- 5 a \$9.75 b \$16.63 c \$0.83 d No answer given
e \$17.46 f \$1746.00

16 - Building a fashion collection

- 1 \$91.96
- 2 a 32 b 48
- 3 120 4 4; 4
- 5 a Size 34 b Size 12
- 6 31% 7 25%
- 8

Men's Size	28	30	32	34	36	38	Total
Number required for 3 buying packs	6	12	18	24	12	6	78
Women's Size	8	10	12	14	16		
Number required for 3 buying packs	6	6	12	9	3		36

- 9 a 78 b 36 c 114
- 10 \$17 100.00
- 11 a \$3 600 b \$1 800 c \$5 400
- 12 32%

17 - Matching tops and bottoms

- 1 20 2 37, 39, 41, 43 inches respectively
- 3 94, 99, 104, 109 cm
- 4

Men's Size	28	30	32	34	36	38	Total
No. required per buying pack	2	4	6	8	4	2	26
No. for 3 buying packs	6	12	18	24	12	6	78
Poplin - Black (core)							
No. for 3 buying packs	6	12	18	24	12	6	78
Poplin - Charcoal (core)							
No. for 2 buying packs	4	8	12	16	8	4	52
Poplin - Taupe (accent)							
No. for 2 buying packs	4	8	12	16	8	4	52
Poplin - white (accent)							
Totals	20	40	60	80	40	20	260

- 5 a 20; 40; 60; 80; 40; 20 b 260
- 6 \$1580; \$3160; \$4740; \$6320; \$3160; \$1580
- 7 \$20 540
- 8

Women's Size	8	10	12	14	16	Total
Total number of each size	20	24	40	34	6	124
Amount received on sales = number x selling price	\$1700	\$2040	\$3400	\$2890	\$510	\$10 540

- 9 \$20 540 + \$10 540 = \$31 080 10 34%

18 - Extending the collection

- 1 17 2 finer 3 50 g
- 4 a 8.75 m² b 437.5 g
- 5 a 100 g b 84 6 \$96.51

- 7 a 70 grams b 930 grams c 2 d 210 grams
- 8 a 19.355 kg b 20 c \$450 d \$13 500.00
- 9 a \$94.50 b \$175.50 c \$8750
- 10 a \$72.90 b \$197.10 c \$9850

19 - Why do these clothes cost so much?

- 1 a \$23.00 b \$46.00 c 4:1
- 2 \$51.00 3 a \$51.00 b \$51.00
- 4 \$102.00 5 1/3 6 200 % 7 \$102.00 8 2/3
- 9 a \$19.00 b 66 2/3 % c 66.89%
- 10 68.75% 11 74.67% 12 \$16.20
- 13 a \$87.50 b 35% c \$2100.00
- 14 a \$36.00 b \$84.00 c (i) \$3132.00 (ii) \$7308.00

20 - Making a profit or a loss

- 1 a \$30 000 b 200 c \$20 000 d \$17 500
e 58.33% f \$2 500

	Calculations in \$
Amount received from sales	\$30 000.00
Cost of Goods Sold (COGS)	\$10 000.00
Return = Sales - COGS	\$20 000.00
Operating Expenses	\$17 500.00
Profit Before Tax	\$2 500.00

	Calculations in \$
Amount received from sales	\$30 000.00
Cost of Goods Sold (COGS)	\$8 500.00
Return = Sales - COGS	\$21 500.00
Operating Expenses	\$17 500.00
Profit Before Tax	\$4 000.00

- 2 \$3000 3 \$17 000
- 4 a \$900 b \$2 100.00 c 7%
- 5

	Calculations in \$
Net sales	\$30 000.00
Cost of Goods Sold (COGS)	\$10 000.00
Gross Margin = Net Sales - COGS	\$20 000.00
Operating Expenses	\$17 000.00
Profit Before Tax	\$3 000.00
Profit after 30% Company Tax	\$2 100.00

21 - UDUDE goes solo

- 1 48 m²
- 2 a 24m² b 1.57m² c 22.43m²; ~47%
d \$23 182.24
- 3 a 10 metres b 62.5% c Jackets: \$156 000.00; Pants: \$78 000.00; Jeans: \$78 000.00; Shirts: \$104 000.00; Knitwear: \$52 000.00; T-shirts: \$26 000.00; Accessories: \$26 000.00
- 4 a \$10 000.00 b \$43 333.33
- 5 a \$26 000.00 b \$173 333.33 c 346 666.67
d 77 826.67 e 15% f \$23 348.00 g \$54 478.67
- 6 a 10.47 % b yes

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